PAPERCLIP

AI AGENT COMPANY OS

COURSE SLIDES · WSQ

Manage AI Agents with Paperclip

WSQ Course Code: TGS-2023036646

TSC: Applications Integration · ICT-DIT-3003-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
Generative AI, large language models, agentic AI systems and applied machine learning.

DELIVERS
WSQ courses on AI, agentic AI, LLM application development, prompt engineering and data science.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

1

Who you are

Your name, your organisation and the role you play in it.

2

What you've used

Your experience with AI agents, automation or command-line tools — if any. No experience needed.

3

What you'd delegate

One task in your work you would hand to a team of AI agents.

4

What you wouldn't

One task you would NOT delegate — and the reason why. We come back to this in Topic 1.

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HOUSEKEEPING

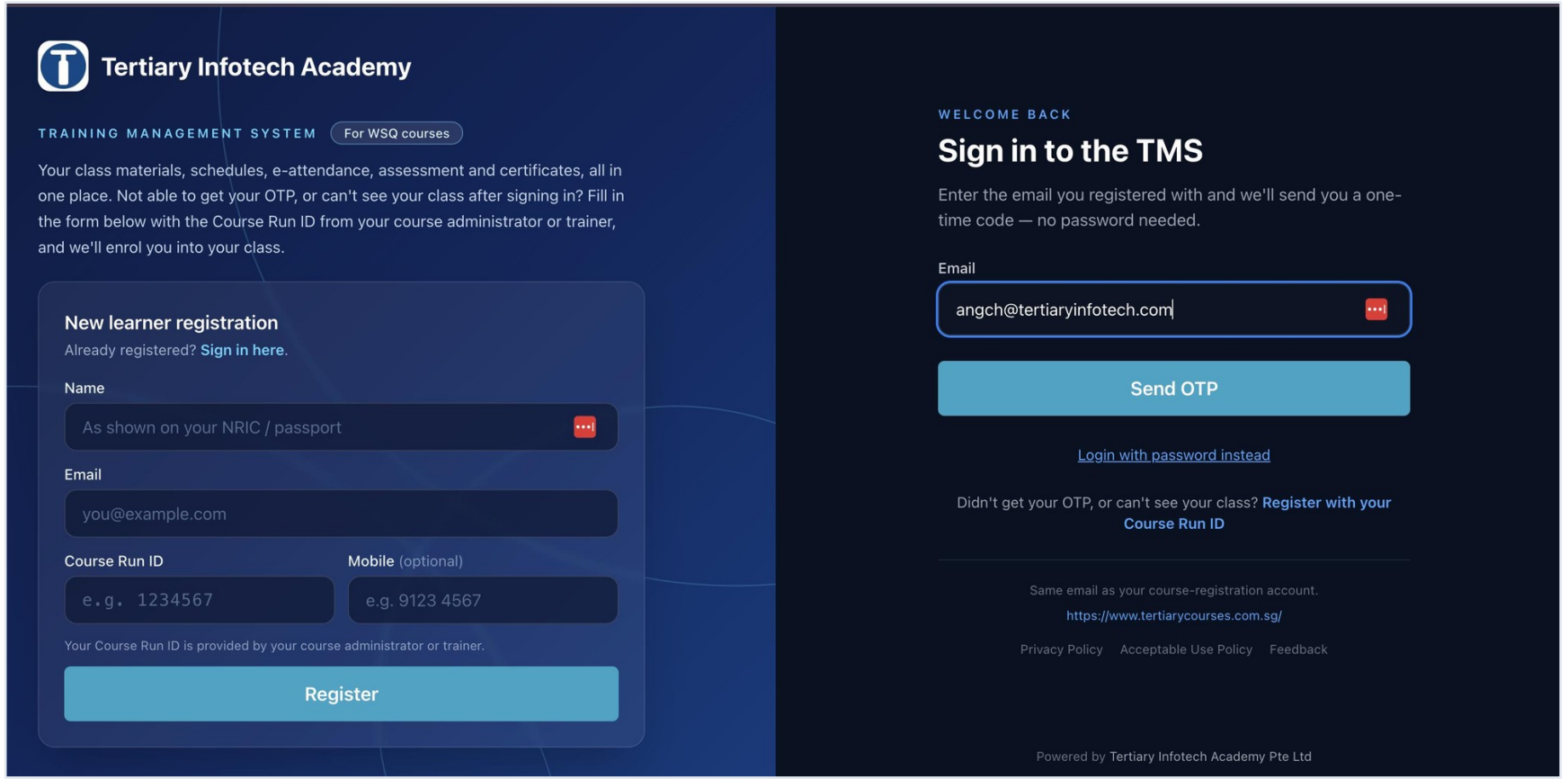
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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LMS / TMS · lms-tms.tertiaryinfotech.com

Download Your Course Material

1

Go to the LMS portal

Open https://lms-tms.tertiaryinfotech.com in your browser.

2

Log in

Sign in with the account e-mail you used to register for this course.

3

Open this course

Select 'Manage AI Agents with Paperclip'.

4

Download

Download the Trainer Slides, Learner Guide and Lesson Plan (PDF).

5

Get the labs

Clone the lab repository from GitHub — the link is on the course page.

6

Keep them open

You may use these materials during the open-book assessment.

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SKILLS FRAMEWORK · TSC

Skills Framework Alignment

1

TSC Title

Applications Integration

2

TSC Code

ICT-DIT-3003-1.1

3

Training hours

14 hours across 2 days

4

Assessment hours

2 hours — Written Assessment + Practical Performance

5

Level

Beginner to Intermediate

6

Mode

Physical, instructor-led with hands-on labs

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SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

Day 1

- Day 1 — Agentic AI foundations, the Paperclip control plane, and building your first goal-aligned AI agent company

– Digital Attendance (AM) · Introductions · Learning Outcomes

– Topic 1: Fundamentals of Agentic AI and Paperclip (Lab 1)

– Lunch Break · Digital Attendance (PM)

– Topic 2: Building AI Agents with Paperclip (Labs 2–3)

– Day 1 recap

Day 2

- Day 2 — Multi-agent operations: decomposition, hand-offs, routines, budgets, approvals and the closed audit loop

– Digital Attendance (AM) · Day 1 recap

– Topic 3: Developing Intelligent Multi-Agent Systems (Labs 4–5)

– Lunch Break · Digital Attendance (PM)

– Topic 4: Deploying and Managing AI Agents (Lab 6)

– TRAQOM Survey · Digital Attendance (Assessment)

– Final Assessment (WA + PP) — 2 hours

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Identify & scan feasibility

Explain the Paperclip control-plane model and judge when a managed multi-agent company earns its coordination cost.

2

LO2 · Integrate & develop

Install Paperclip, create a goal-aligned company, build the reporting tree and connect agent runtimes, skills and workspaces safely.

3

LO3 · Test & verify

Decompose a mission into issues, run bounded heartbeats and verify hand-offs through runs, comments, artifacts and the activity log.

4

LO4 · Resolve & modify

Diagnose adapter, budget and routine failures, and modify governance, approvals and cost boundaries to keep the operation safe.

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FOUR TOPICS · 6 HANDS-ON LABS

Course Outline

1

Topic 01 — Fundamentals of Agentic AI and Paperclip

Agentic AI foundations · the control-plane model · the twelve subsystems · feasibility and when NOT to use multi-agent

2

Topic 02 — Building AI Agents with Paperclip

Install and verify · company and mission · the CEO · org chart · runtime adapters · workspaces, instructions and secrets

3

Topic 03 — Developing Intelligent Multi-Agent Systems

Projects, goals and lineage · issue decomposition · assignment and hand-off · sequential specialists · verifying between disparate components

4

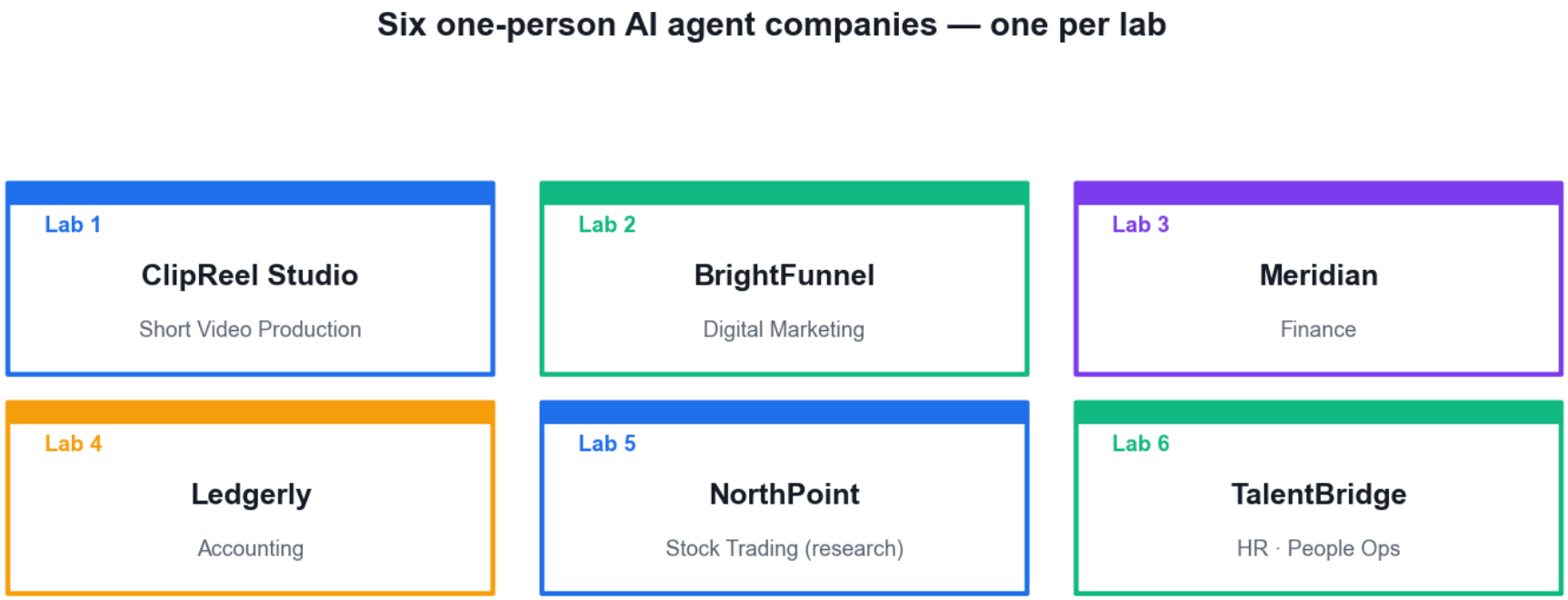
Topic 04 — Deploying and Managing AI Agents

Routines and schedules · layered budgets · approvals and audit · operator intervention · deployment, portability and troubleshooting

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THE COURSE SIMULATION

Six One-Person AI Agent Companies



Every data set is synthetic — built for training, never real company, market or candidate data.

Each lab builds a different one-person company — so you see the same control model applied to six very different businesses.

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BEFORE YOU START

Briefing for Assessment

1

Do · Clear your desk

Place phones and other materials under the table or on the floor.

2

Don't · No recording

No photos or recording of assessment scripts.

3

Don't · No discussion

Work individually — no discussion during the assessment.

4

Do · Black or blue pen

Use a black or blue pen for hard-copy assessments.

5

Don't · No correction fluid

No liquid paper or correction tape may be used.

6

Do · Stop on time

Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

1

Written Assessment (WA)

Short-Answer Questions (SAQ) · 5 questions · 1 hour · open book. Tests the underpinning knowledge (K1–K5).

2

Practical Performance (PP)

4 hands-on Paperclip tasks · 1 hour · open book. Tests the ability you built in the labs (A1–A4).

3

Open book

You may use the course slides, the Learner Guide and approved materials only.

4

Eligibility

A minimum of 75% attendance is required to be eligible for assessment and funding.

5

Result

You are assessed as Competent (C) or Not Yet Competent (NYC) on each instrument.

6

Appeals

An appeal process is available if you wish to contest an assessment outcome.

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ON ASSESSMENT DAY

Assessment Flow

1

TRAQOM survey

Scan the QR code on the LMS

2

Digital attendance

Scan the SSG assessment QR

3

Sit WA then PP

Open book · 1 hour each

4

Submit on the LMS

Upload your completed answers

5

Sign the record

Sign the Assessment Summary Record

REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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EXAMS.TERTIARYINFOTECH.COM

Practice Exam

1

Practise before the real thing

Try the practice questions at https://exams.tertiaryinfotech.com before the final assessment.

2

Same format

Short-answer knowledge questions in the same style as the Written Assessment.

3

Self-marked

Use it to find the topics you need to revisit, not to score yourself.

4

Free and unlimited

Attempt it as many times as you like during the course.

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WSQ FUNDING

Criteria for Funding

1

Attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

Assessment

Complete both assessment components and be assessed as 'Competent'.

3

Digital attendance

Scan the SSG QR code for AM, PM and Assessment on every training day.

4

TRAQOM survey

Complete the mandatory TRAQOM course feedback survey on the LMS.

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LAB ENVIRONMENT

Set Up Before We Start

1

Node.js 20 or later

Paperclip requires Node 20+. Check with node --version.

2

A normal user shell

Never run onboarding from an elevated Administrator or root shell.

3

An agent runtime

One trainer-approved CLI runtime (e.g. Claude Code, Codex, Gemini CLI).

4

Git

Used to baseline each agent workspace so every change is a readable diff.

5

Lab repository

Clone the course labs — the link is on your LMS course page.

6

Synthetic data only

All six company data sets are fabricated for training. Never load real company, market or candidate data.

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CORE CONCEPTS

From a Single Agent to a Managed Agent Company

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THE SIX INGREDIENTS

What Makes a System 'Agentic'?

1

Reasoning

The model plans a sequence of steps instead of answering in one shot.

2

Tool use

It calls functions, CLIs and APIs to fetch live data or take real actions.

3

Memory

State carries context across turns — and, in Paperclip, across heartbeats.

4

Autonomy

It decides which tool to use and when the work is actually finished.

5

Collaboration

Several specialised agents divide the work and hand off between themselves.

6

Guardrails

Policies, budgets and approvals constrain what the agent may do.

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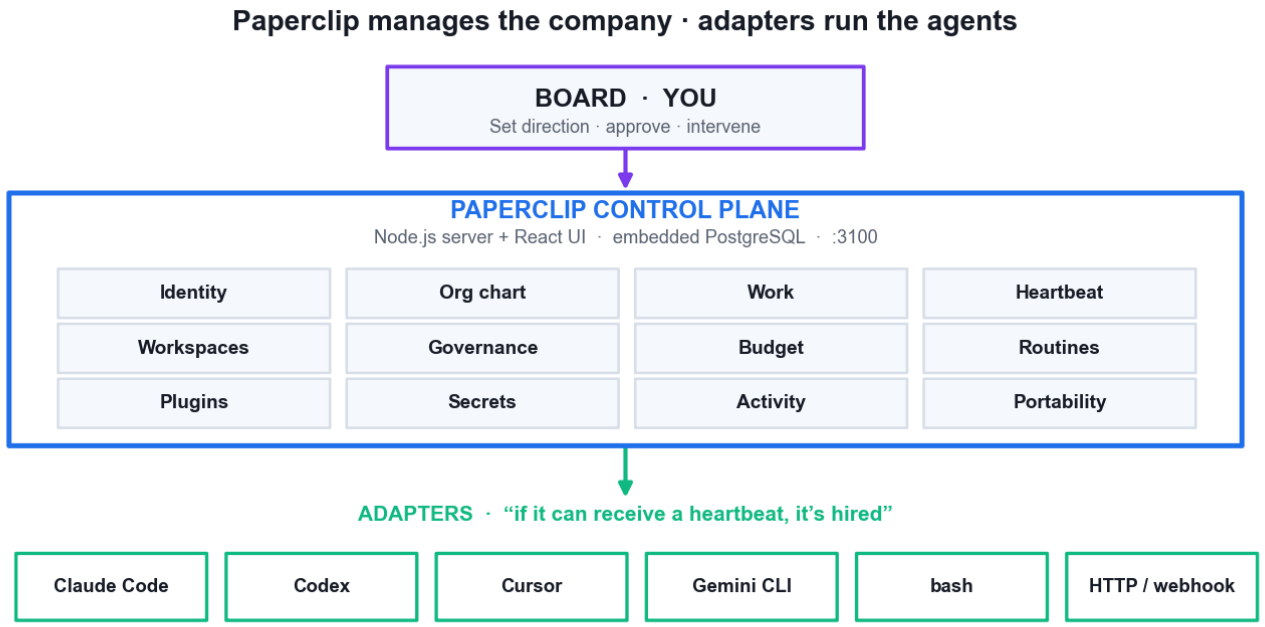
WHY A CONTROL PLANE

An LLM answers. An agent acts. A company coordinates.

Paperclip is the third layer — it does not make your agents smarter, it makes a team of them accountable.

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Paperclip Control-Plane Architecture



The runtimes do the reasoning and the tool use. Paperclip never replaces them — it schedules, budgets and audits them.

Board · you

You set direction, approve governed actions and intervene. Authority never leaves the human.

The control plane

Stores companies, org charts, issues, budgets, secrets and the audit trail. Node.js + React on port 3100.

The adapters

Claude Code, Codex, Cursor, Gemini CLI, bash or an HTTP bot — these do the actual reasoning.

The separation

Organisational authority, runtime capability and verification evidence are kept deliberately apart.

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LO1 · SCOPE THE TOOL CORRECTLY

What Paperclip Is — and Is Not

1

IS · an organisational control plane

Org charts, issues, budgets, approvals and a durable audit trail for a team of agents.

2

IS · a runtime-agnostic scheduler

'If it can receive a heartbeat, it's hired.' It schedules and budgets agents without replacing them.

3

IS NOT · an agent framework

It does not dictate how your agent reasons, prompts or calls tools. That is the adapter's job.

4

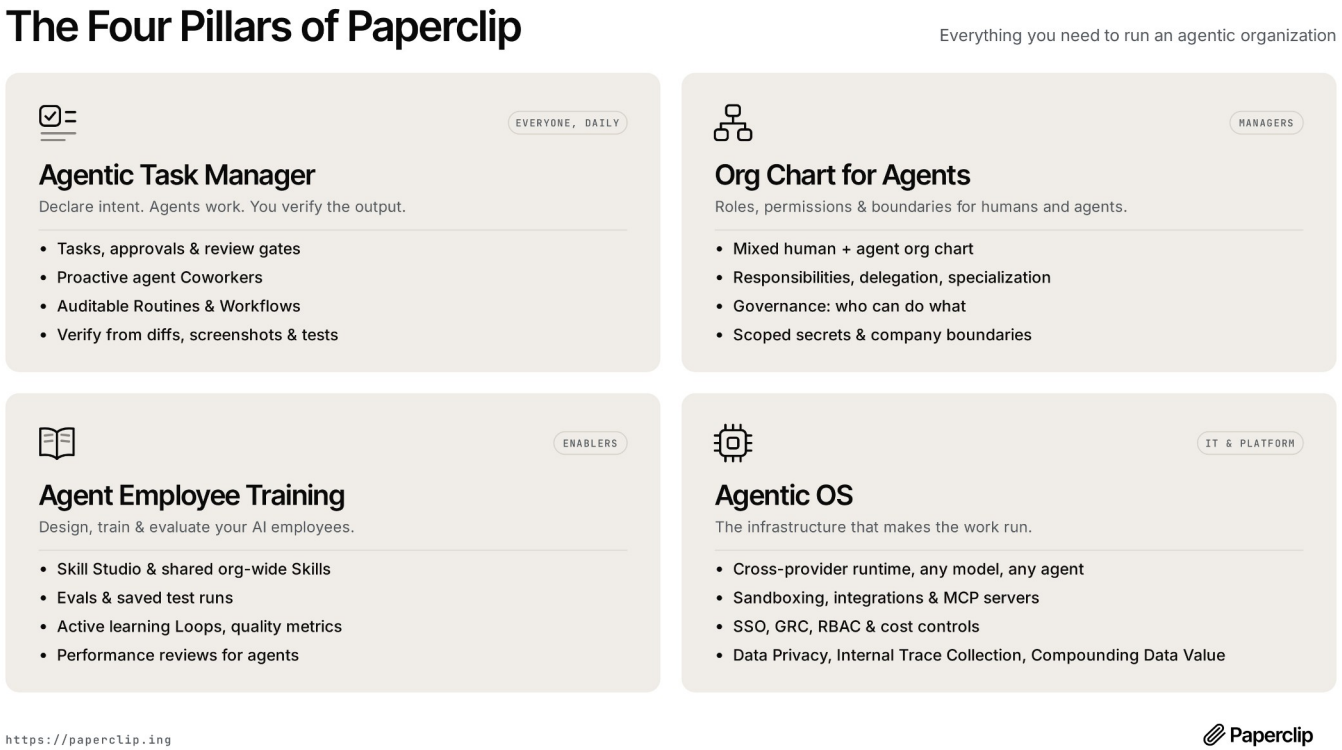
IS NOT · a chatbot or workflow builder

No drag-and-drop canvas, no prompt manager, and not designed for a single-agent use case.

Choosing Paperclip is an organisational decision, not a modelling decision.

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The Four Pillars of Paperclip



Agentic task manager · org chart for agents · agent employee training · agentic OS. Source: the official Paperclip project.

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The Twelve Subsystems

1. **Identity & org chart**
   Who exists, who reports to whom, and what each role may do.

2. **Work & heartbeat**
   Issues with atomic checkout, woken by bounded execution bursts.

3. **Workspaces & runtime**
   Isolated roots (git worktrees) and the adapter that runs there.

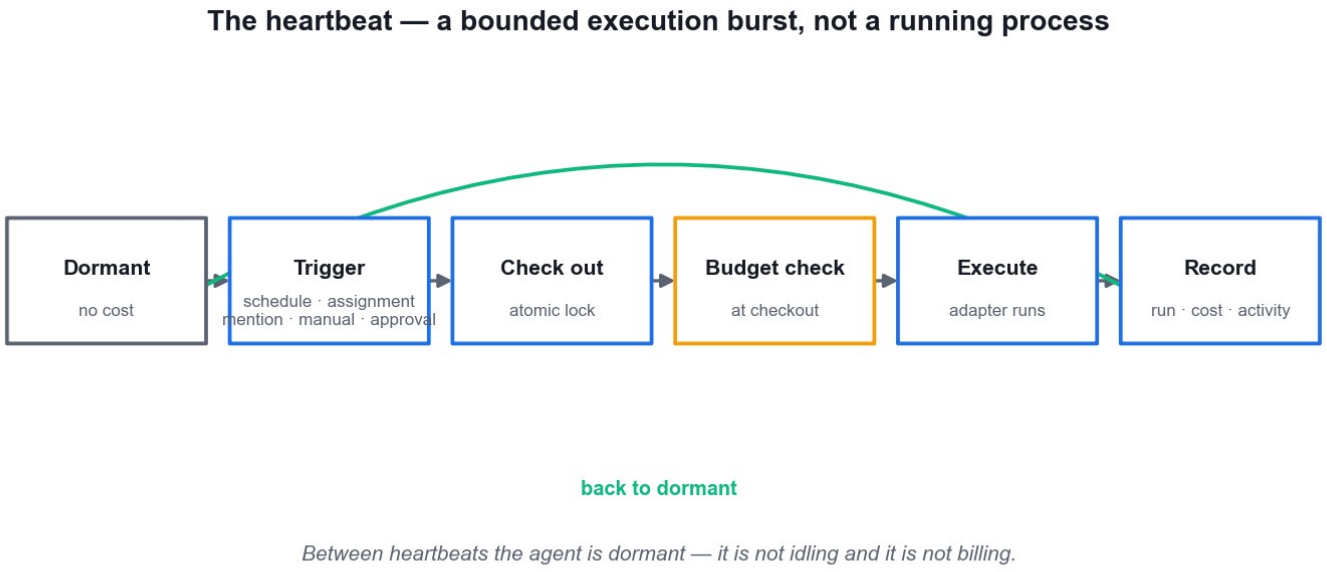
4. **Governance & budget**
   Approval gates and caps enforced at checkout, not at billing.

5. **Routines & plugins**
   Scheduled recurring work, and the skills library agents draw on.

6. **Secrets, activity & portability**
   Scoped injection, a durable audit log, and export/import of a whole company.

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The Heartbeat — Bounded Execution, Not a Running Process



Dormant by default

Between heartbeats the agent is not idling and not billing.

Five triggers

A schedule/interval tick, a task assignment, an @-mention, a manual Run Heartbeat, or an approval being resolved.

Atomic checkout

An execution lock means two heartbeats can never double-work the same issue.

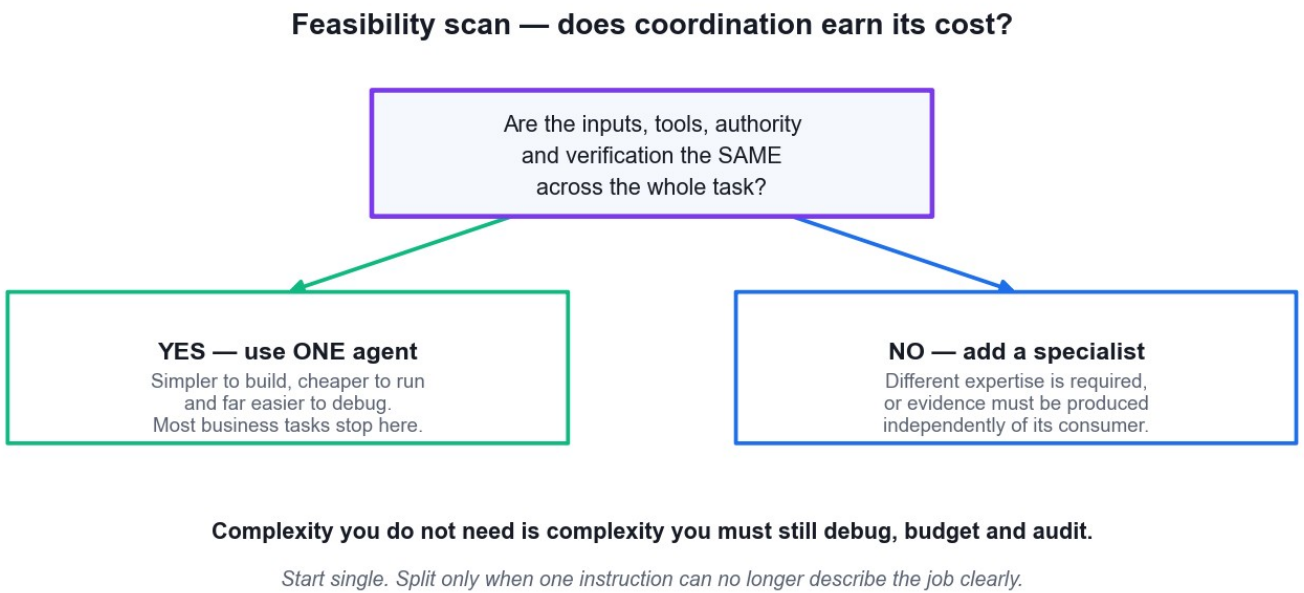
Budget checked first

The cap is tested at checkout — a runaway loop stops at the ceiling.

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LO1 · THE MOST IMPORTANT DECISION IN THE COURSE

Feasibility Scan — Does Coordination Earn Its Cost?



Complexity you do not need is complexity you must still debug, budget and audit.

Start single. Split only when one instruction can no longer describe the job clearly.

Use ONE agent

When inputs, tools, authority and verification are the same across the whole task.

Add a specialist

Only when different expertise is required, or evidence must be produced independently.

The real cost

Every extra agent is another workspace, budget, instruction and audit trail to maintain.

Start single

Split only when one instruction can no longer describe the job clearly.

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One Agent vs a Paperclip Agent Company

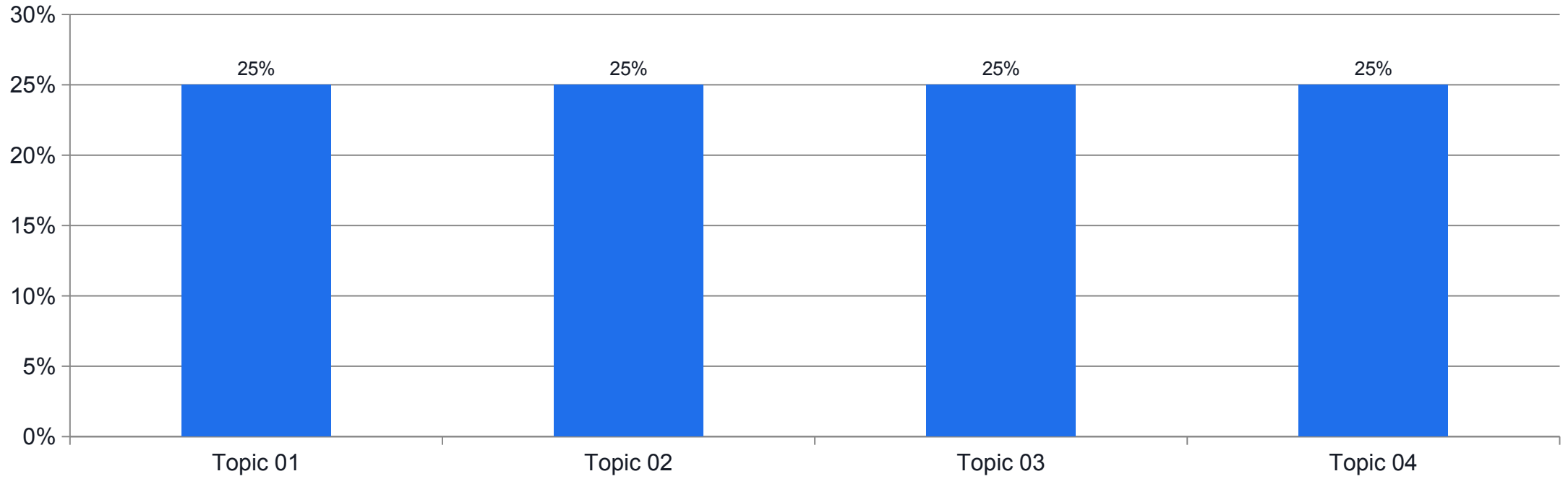
Dimension	One agent, run by hand	A Paperclip agent company
Direction	Restated in every prompt	A mission that flows down as goal lineage
Memory	Ends with the session	Persists across heartbeats in the control plane
Cost	Discovered on the invoice	Capped at checkout per company, project and agent
Authority	Implicit — whatever you allow	Explicit — org chart, roles and approvals
Evidence	Scrollback you must trust	Runs, comments, diffs and a durable activity log
Main risk	Context saturation	Coordination overhead you did not need

WHEN IT MATTERS The right answer is often still ONE agent. Reach for the company when the work has genuinely separable roles that must each be evidenced.

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LESSON PLAN · TIME ALLOCATION

Where Course Time Goes — Topic Weighting



WHAT THE DATA SHOWS

The four topics carry equal weighting — but Topics 3 and 4 are where the assessable practical skill sits, because that is where governance and verification are actually exercised.

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01

TOPIC 01

Fundamentals of Agentic AI and Paperclip

Agentic AI foundations · the control-plane model · the twelve subsystems · feasibility and when NOT to use multi-agent

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TOPIC WEIGHTING 25%

Key Concepts — Fundamentals of Agentic AI and Paperclip

1. Agentic AI adds reasoning, tool use, memory and autonomy on top of a language model, so the system completes work rather than only producing text.

2. Paperclip is an organisational control plane — a Node.js server with a React UI that coordinates agents, goals, work, cost and governance. It is not an agent framework and it does not perform the reasoning itself.

3. The defining separation: Paperclip manages the company; adapters run the agents. Organisational authority, runtime capability and verification evidence stay apart.

4. A company is the boundary for one mission, its human members, AI employees, projects, issues, budget and activity history. One deployment holds many companies.

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TOPIC WEIGHTING 25%

Key Concepts — Fundamentals of Agentic AI and Paperclip (continued)

1. The four pillars are the agentic task manager, the org chart for agents, agent employee training (skills and evaluations) and the agentic OS (runtime, sandboxing, MCP, RBAC and cost control).

2. Twelve subsystems make up the platform: identity, org chart, work, heartbeat, workspaces, governance, budget, routines, plugins, secrets, activity and portability.

3. A heartbeat is a bounded execution window with FIVE triggers: a schedule/interval tick, a task assignment, an @-mention in a comment, a manual Run Heartbeat, or an approval being resolved. Between heartbeats the agent is dormant — not idling and not billing.

4. Work is checked out atomically with an execution lock, so two heartbeats can never double-work the same issue or race past a budget ceiling.

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TOPIC WEIGHTING 25%

Key Concepts — Fundamentals of Agentic AI and Paperclip (continued)

1. 'If it can receive a heartbeat, it's hired' — adapter type keys include claude_local, codex_local, gemini_local, opencode_local, cursor, pi_local and hermes_local, plus process, http and openclaw_gateway.

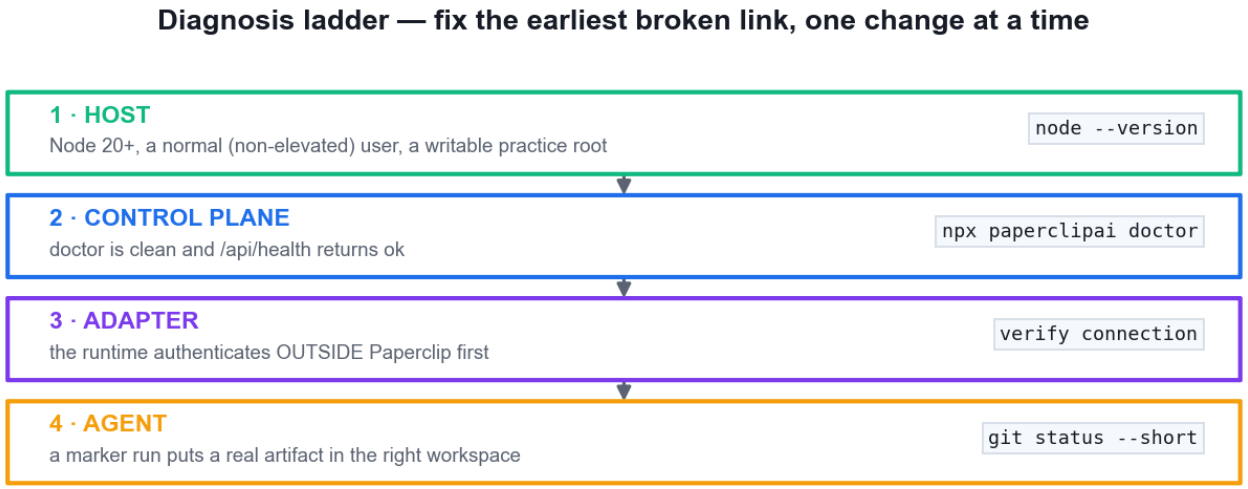
2. Feasibility comes before architecture: use one agent when the inputs, tools, authority and verification are the same across the whole task.

3. Add a specialist only when different expertise is genuinely required, or when evidence must be produced independently of the agent that will use it.

4. Installation is ready only when Node.js, the Paperclip configuration, the embedded database, server health, the browser UI and the doctor diagnostic all agree.

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Diagnosis Ladder — Fix the Earliest Broken Link



Never change several settings at once — you lose the ability to say what actually fixed it.

Host first

Node 20+, a normal non-elevated user, a writable practice root.

Then the control plane

doctor is clean and /api/health returns ok.

Then the adapter

The runtime must authenticate OUTSIDE Paperclip first.

Then the agent

A marker run puts a real artifact in the right workspace.

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Installation Is Ready Only When Everything Agrees

1. **Node.js**
   Major version 20 or later — the most common cause of a failed onboard.

2. **Configuration & database**
   Written by onboard; the embedded PostgreSQL starts as the normal user.

3. **Server health**
   GET /api/health returns an ok status.

4. **Browser UI**
   http://localhost:3100 loads without a connection error.

5. **Doctor**
   npx paperclipai doctor completes with no blocking error.

6. **Evidence**
   All of the above recorded — with no credential ever captured.

A green badge is not evidence. An observable artifact is.

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The Four Additions That Make an Agent

Addition	What it provides	What breaks without it
Reasoning / planning	Decomposes a goal into steps before acting, and re-plans when a step fails.	Answers in one shot; cannot recover from a wrong first move.
Tool use	Calls functions, CLIs and APIs to read live data and take real action.	The system can only talk about the work; it cannot do it.
Memory / state	Context carries across turns and across separate execution windows.	Every wake-up starts from nothing and repeats work already done.
Autonomy	Decides which tool, in which order, and when it is finished.	A human drives every step — a chatbot with extra latency.

WHEN IT MATTERS Add a fifth ingredient — several agents dividing the work — and you have a multi-agent system. That is where the interesting failures start.

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Why Coordination Is the Hard Part

1

DIRECTION

Each agent must know not only what to do but WHY — or it optimises the wrong thing very efficiently.

2

AUTHORITY

Who may hire, spend and ship. Left implicit it defaults to 'whatever the agent decided to try'.

3

ISOLATION

Two agents writing to the same place corrupts both pieces of work and destroys attribution.

4

COST

An agent in a retry loop does not get tired. Without a ceiling you discover the problem on the invoice.

5

EVIDENCE

'The agent says it is done' is not 'it is done'. Someone must check later, without taking anyone's word.

6

THE MAPPING

These five concerns are exactly what the twelve subsystems address — map each subsystem back to one.

Running one agent is a prompting problem. Running five is an organisational one.

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LO1 · THE ASSESSMENT USES THIS VOCABULARY

The Official Vocabulary — Learn It Verbatim

Term	The documentation's own definition
Company	A self-contained AI organisation with a single goal, a team of agents, a task board, and a budget.
Agent	An AI employee — a configured role with a specific job, a manager, and a monthly budget.
Task (issue)	A unit of work — like a ticket on a project board, except it is assigned to an AI agent, not a human.
Heartbeat	A scheduled execution window — when an agent wakes up, checks its inbox, and does work.
Adapter	The connection between Paperclip and the AI system that actually runs your agent.
Budget	A monthly spending cap that Paperclip enforces automatically so your agents never overspend.
Approval	A governance gate — a checkpoint where Paperclip pauses and asks you to sign off before an action proceeds.

WHEN IT MATTERS Notice what is absent: nothing about prompting, model selection or how an agent reasons. That omission is deliberate.

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LO1 · THREE LAYERS, THREE DIFFERENT PROBLEMS

LLM vs Agent vs Agent Company

Dimension	A language model	A single agent	A Paperclip company
What it does	Answers	Acts	Coordinates several actors
Memory	None beyond the prompt	Within one session	Across heartbeats, in the control plane
Authority	None	Whatever you allow implicitly	Explicit — org chart, roles and approvals
Cost visibility	Per request	Discovered on the invoice	Capped at checkout by company, project and agent
Evidence	The reply text	Scrollback you must trust	Runs, comments, diffs and a durable activity log

WHEN IT MATTERS Paperclip is the third layer. It does not make your agents smarter — it makes a team of them accountable.

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What Paperclip Is NOT

1

NOT an agent framework

It does not dictate how your agent reasons, prompts or calls tools. That is the adapter's job.

2

NOT a chatbot

There is no conversational surface for you to talk to a model through.

3

NOT a workflow builder

There is no drag-and-drop canvas of nodes and edges to wire together.

4

NOT a prompt manager

Prompts live with the runtime and the skill, not in a prompt library inside Paperclip.

5

NOT for a single agent

One agent does not need an org chart — the coordination overhead would be pure cost.

6

SO WHAT IS IT?

An organisational control plane: Node.js plus React on port 3100, backed by PostgreSQL on port 54329.

Paperclip manages the company. Adapters run the agents.

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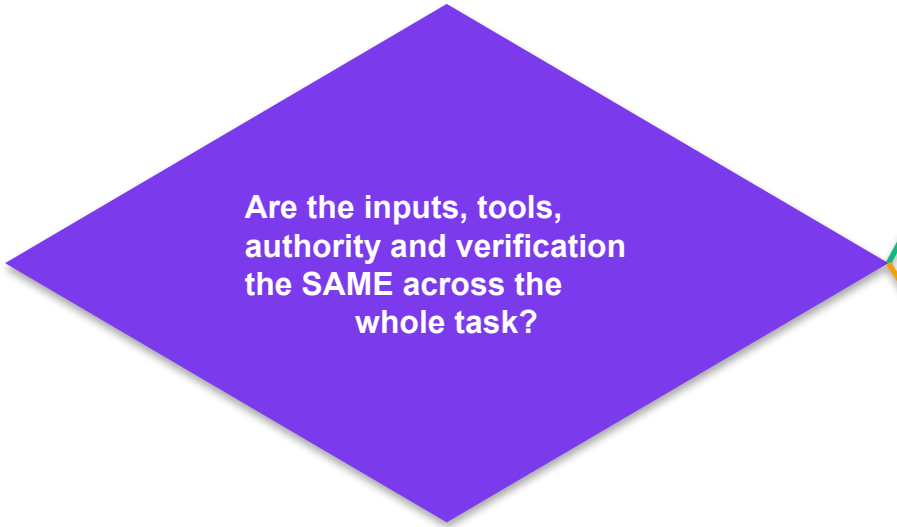
The Feasibility Scan — One Agent or a Team?

Factor	Points to ONE agent	Points to a TEAM
Separability	One instruction still describes the job clearly.	The instruction has become a list of unrelated responsibilities.
Expertise	One skill set covers it.	Genuinely different expertise is required for different parts.
Independent evidence	The doer can reasonably check their own work.	Evidence must be produced INDEPENDENTLY of whoever consumes it.
Observability	Inputs and outputs are already observable.	The same — a prerequisite either way, not a reason to split.
Cost of coordination	Low volume, short task.	The work recurs often enough to amortise the setup.

WHEN IT MATTERS The test is one question: are the INPUTS, TOOLS, AUTHORITY and VERIFICATION the same across the whole task? If yes, use one agent.

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Should This Work Become an Agent Company?



YES — stay with ONE agent

One instruction still describes the job clearly. Splitting adds nothing but surface area to debug, budget and audit.

NO — build the company

Different expertise, or evidence must be produced independently of whoever will consume it. Split into roles, one transformation each.

Sanity check before you build anything: write the mission as one sentence. If you cannot, you do not understand the work well enough to delegate it.

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LO1 · KNOW WHERE TO LOOK WHEN SOMETHING MISBEHAVES

Twelve Subsystems — Who Owns Which Problem?

Subsystem	The question it answers
Identity · Org chart	Who is this, who do they report to, and what may they do?
Work · Heartbeat	What needs doing, who owns it now, and when does an agent wake?
Workspaces	Where does this agent's work live, and who else can touch it?
Governance · Budget	Which actions need a human decision, and what is the ceiling?
Routines · Plugins	What recurs unasked, and how do we extend without forking?
Secrets	Where do credentials live, and how do they reach a run without leaking?
Activity · Portability	What happened, in order and durably — and how do we move it?

WHEN IT MATTERS You do not need to recite the twelve. You should be able to say which one owns a given problem — that is how you know where to look.

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WHAT YOU'LL DO

Hands-On Labs — Fundamentals of Agentic AI and Paperclip

Lab 1

- Install Paperclip and Prove the Control Plane (ClipReel Studio)
- ClipReel Studio (Short Video Production)

—

- —

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LAB 1 · HANDS-ON

Install Paperclip and Prove the Control Plane (ClipReel Studio)

SIMULATED COMPANY · ClipReel Studio (Short Video Production)

You prepare a dedicated synthetic practice root for ClipReel Studio, a one-person short-video production company, verify the supported Node.js runtime, onboard Paperclip as a normal (non-elevated) user, and then inspect BOTH the built-in diagnostic and the live health endpoint. Proving the control plane before you hire any agent matters because every later adapter, budget or heartbeat failure is far easier to isolate when the platform itself has a recorded clean baseline.

YOU'LL BUILD

A running Paperclip instance at http://localhost:3100 plus evidence/lab01-installation.md recording the Node/npm versions, the doctor summary and the /api/health response — with no credentials captured.

MAPS TO

LO1: Explain the control-plane / runtime boundary and verify a healthy local Paperclip installation with observable evidence.

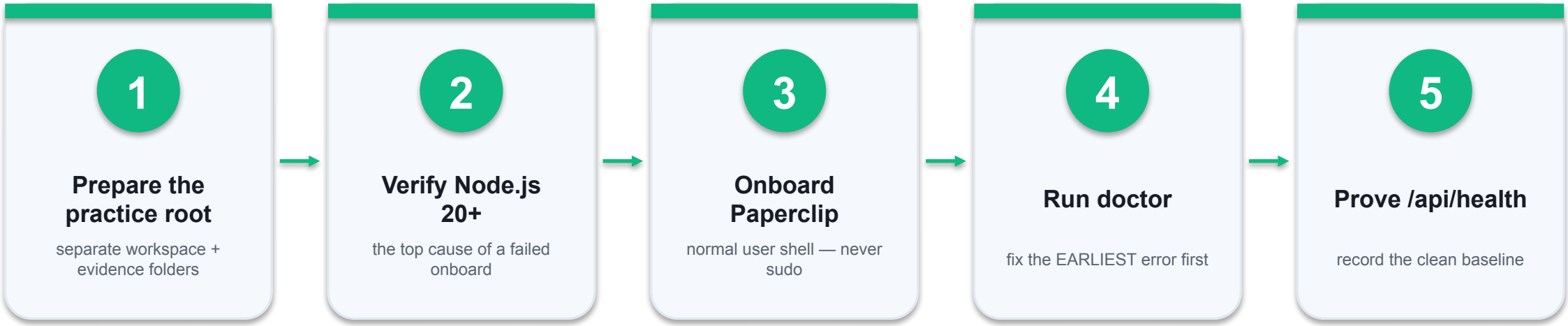
Tools: Node.js 20+, npm/npx, Paperclip CLI, browser, terminal or PowerShell

Full step-by-step procedure: Learner Guide, Lab 1 · labs/ folder

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LAB 1 · PROCESS MAP

How Lab 1 Runs



THE POINT

LO1: Explain the control-plane / runtime boundary and verify a healthy local Paperclip installation with observable evidence.

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LAB 1 · VERIFY

Install Paperclip and Prove the Control Plane (ClipReel Studio)

✅ Expected result

npx paperclipai doctor completes with no unresolved blocking error, GET http://localhost:3100/api/health returns status ok, the browser loads the Paperclip UI, and evidence/lab01-installation.md records all three checks with no credential or personal identifier.

IF IT DOESN'T WORK

Embedded PostgreSQL refuses to start
Confirm the shell is NOT elevated. Stop the process and re-run onboarding as the normal user who owns the practice folder.

The browser cannot reach port 3100
Run npx paperclipai doctor, then npx paperclipai run, and query /api/health again BEFORE touching any firewall setting.

npx resolves against a private registry
Retry with npx --registry https://registry.npmjs.org paperclipai onboard --yes and review the global npm registry configuration afterwards.

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LAB 1 · TROUBLESHOOTING

Install Paperclip and Prove the Control Plane (ClipReel Studio)

⚠ Embedded PostgreSQL refuses to start

Confirm the shell is NOT elevated. Stop the process and re-run onboarding as the normal user who owns the practice folder.

⚠ The browser cannot reach port 3100

Run npx paperclipai doctor, then npx paperclipai run, and query /api/health again BEFORE touching any firewall setting.

⚠ npx resolves against a private registry

Retry with npx --registry https://registry.npmjs.org paperclipai onboard --yes and review the global npm registry configuration afterwards.

CHALLENGE

Run npx paperclipai env and record ONLY the non-sensitive instance name, bind mode and port in your evidence note. Do not paste the full output.

REFLECTION

Why does a healthy Paperclip API prove the control plane but say nothing about whether a Claude Code, Codex, Gemini or Cursor runtime is ready?

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TOPIC RECAP

Recap — Fundamentals of Agentic AI and Paperclip

- You can now: LO1: Explain the control-plane / runtime boundary and verify a healthy local Paperclip installation with observable evidence.
- Agentic AI adds reasoning, tool use, memory and autonomy on top of a language model, so the system completes work rather than only producing text.
- Paperclip is an organisational control plane — a Node.js server with a React UI that coordinates agents, goals, work, cost and governance. It is not an agent framework and it does not perform the reasoning itself.
- The defining separation: Paperclip manages the company; adapters run the agents. Organisational authority, runtime capability and verification evidence stay apart.

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02

TOPIC 02

Building AI Agents with Paperclip

Install and verify · company and mission · the CEO · org chart · runtime adapters · workspaces, instructions and secrets

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TOPIC WEIGHTING 25%

Key Concepts — Building AI Agents with Paperclip

1. Onboarding with paperclipai onboard --yes downloads Paperclip, writes the configuration, initialises the embedded PostgreSQL and serves the UI on port 3100.

2. Deployment mode is a security decision: local_trusted is the default and requires no login; authenticated+private and authenticated+public widen reach and require real authentication.

3. The onboarding wizard captures the company name, the mission statement and the team lead, then verifies connectivity to your chosen local agent runtime.

4. A mission must be measurable. 'Grow the channel' cannot be verified; 'publish 8 short videos per month at ≥5% average watch-through' can.

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TOPIC WEIGHTING 25%

Key Concepts — Building AI Agents with Paperclip (continued)

1. Paperclip enforces a strict hierarchy: every agent reports to exactly ONE manager, and the CEO is the only agent with no manager. The CEO owns goal alignment, decomposition and final synthesis, and may propose further hires under approval.

2. Every agent carries a role, a manager, an adapter, an instruction file, a heartbeat and an individual budget. Those six fields are the agent's whole contract.

3. A role instruction states purpose, allowed inputs, the output schema, explicit exclusions and the escalation path — not a personality.

4. Each agent gets a dedicated workspace: a separate local root for one role's files and evidence, isolated with git worktrees so changes are attributable.

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TOPIC WEIGHTING 25%

Key Concepts — Building AI Agents with Paperclip (continued)

1. Secrets live in the runtime or in Paperclip's secret store with scoped injection at heartbeat time. They never appear in prompts, lab notes or Git.

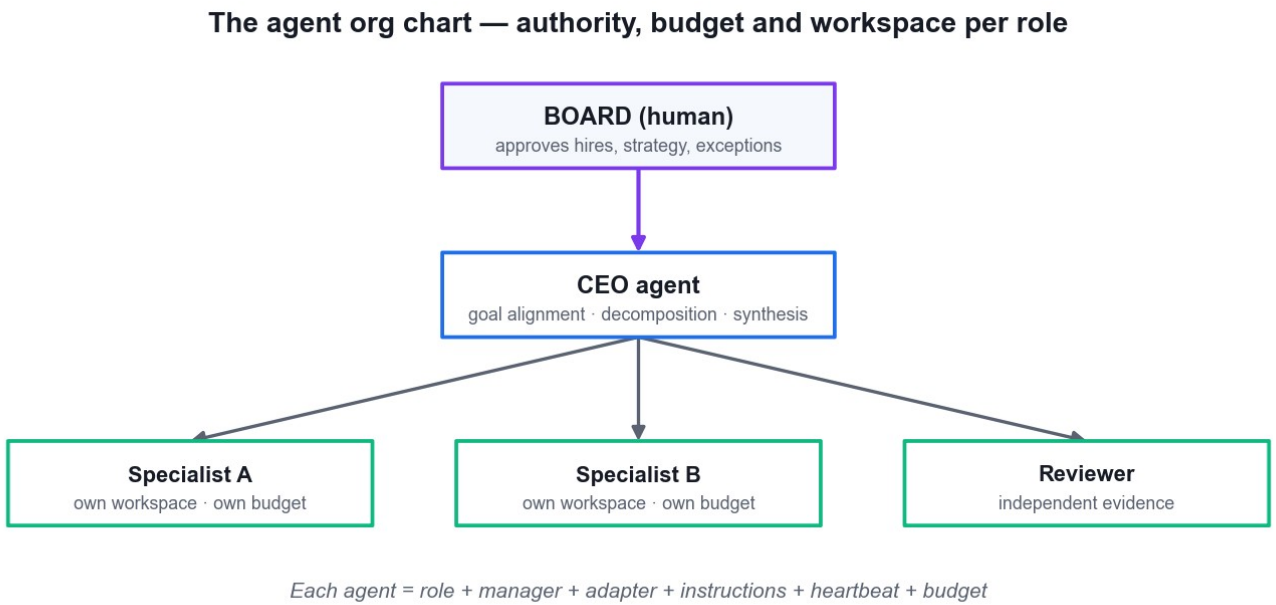
2. Skills are reusable agent capabilities imported from a path or URL into an org-wide library, then attached to specific agents — capability without retraining.

3. Adapter readiness is proved by a marker run, not by the presence of configuration: an agent that cannot produce an observable artifact is not connected.

4. Diagnose along a ladder — host, control plane, adapter, agent — and fix the earliest broken link before changing anything downstream.

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The Agent Org Chart



Board approves

Hires, strategy and material exceptions stay with the human.

The CEO owns alignment

Goal alignment, decomposition and final synthesis.

Specialists own transformations

One distinct job, one authority boundary, one deliverable.

Six fields per agent

Role · manager · adapter · instructions · heartbeat · budget.

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A Role Instruction Is a Contract, Not a Personality

Section	What it states	Why it matters
Purpose	The single transformation this role owns	Prevents scope drift
Allowed inputs	Which files, data and sources may be read	Confines the blast radius
Output schema	The artifact, its path and its required fields	Makes acceptance objective
Exclusions	What this role must never do	The most valuable section
Escalation	When to stop and ask the board	Keeps authority explicit

WHEN IT MATTERS If two roles could both plausibly do the task, your instructions are not finished.

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Workspaces, Secrets and Skills

1

Dedicated workspace

One local root per role, isolated with git worktrees so every change is attributable to exactly one agent.

2

Git baseline

Commit an empty baseline so the later diff means what you think it means.

3

Secrets stay out

Credentials live in the runtime or the Paperclip secret store, injected at heartbeat time — never in prompts, notes or Git.

4

Skills are capability

Import a skill from a path or URL into the org-wide library, then attach it to a specific agent.

5

Adapters bridge capability

Match role complexity, session needs and tool access to the adapter you choose.

6

Prove it with a marker run

Configuration that has never produced an artifact is not a connection.

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The Six Fields That Define Every Agent

Field	What it controls	Get it wrong and
Role	What this agent is for — the single transformation it owns.	Scope drifts into a neighbouring role's work.
Manager	Its reporting line. Exactly one, except for the CEO.	Accountability has no shape; nobody owns the outcome.
Adapter	Which runtime actually executes its work.	The agent is configured but can never run.
Instructions	Its contract: purpose, inputs, output schema, exclusions, escalation.	Acceptance becomes a matter of opinion.
Heartbeat	When it wakes, and on which triggers.	It either never wakes, or wakes and bills when idle.
Budget	Its individual monthly ceiling.	A retry loop spends until someone reads the invoice.

WHEN IT MATTERS Paperclip enforces a strict hierarchy: every agent reports to exactly ONE manager, and the CEO is the only agent with no manager.

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The Five Instruction Sections and Why Each Matters

Section	States	Why it matters
PURPOSE	The single transformation this role owns	Prevents scope drift into a neighbouring role's work.
ALLOWED INPUTS	Which files, data and sources may be read	Confines the blast radius when something goes wrong.
OUTPUT SCHEMA	The artifact, its path, and its required fields	Makes acceptance objective rather than a matter of opinion.
EXCLUSIONS	What this role must NEVER do	The most valuable section — and the one people leave out.
ESCALATION	When to stop and ask the board	Keeps human authority explicit instead of assumed.

WHEN IT MATTERS Tone does not constrain behaviour; structure does. If two roles could each plausibly claim the same task, your instructions are not finished.

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LO2 · FROM THE OFFICIAL DOCUMENTATION

Suggested Starting Budgets by Seniority

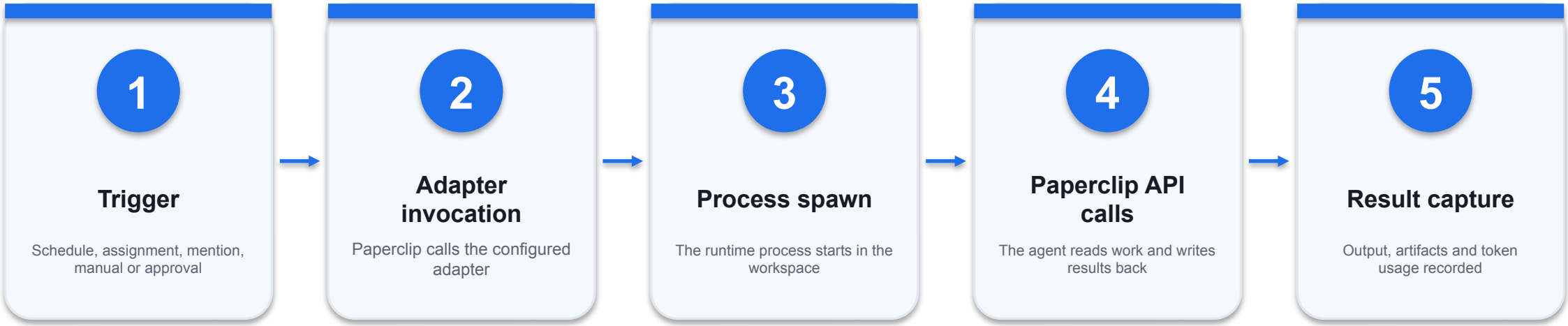
Role	Suggested monthly budget	The reasoning
CEO	$30–50 per month	Owns alignment, decomposition and synthesis — it wakes on more events than anyone else.
Manager (CTO or CMO)	$20–40 per month	Coordinates a sub-team, so it runs more often than a worker but less than the CEO.
Worker agent	$10–25 per month	One transformation, run on assignment. The cheapest role to cap tightly.
Company (total)	Set BEFORE the first hire	Enforcement happens at checkout, so the cap must exist before anyone can check out work.

WHEN IT MATTERS These are a sanity check on your own numbers, not a rule. A cap you would be unhappy to actually pay is set too high.

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LO2 · ALMOST EVERY FAULT IS 'WHICH STEP FAILED?'

The Six-Step Heartbeat Execution Arc



STEP 6 · RUN RECORD STORAGE

The run, its cost and its activity entries are persisted. This rarely fails — but if it does, the audit trail has a gap you must explain.

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Where Each Heartbeat Step Fails

Step	The usual failure	First thing to check
1 · Trigger	Nothing assigned and the interval is off (the default)	The assignee. Creating an issue does not wake anyone.
2 · Adapter invocation	Adapter misconfigured, or runtime not authenticated	Run the runtime's own CLI directly and sign in.
3 · Process spawn	The binary is missing, or the sandbox cannot start	The runtime install, outside Paperclip.
4 · API calls	Checkout conflict (409), or the cross-issue write limit of 20	Whether another agent already holds the issue.
5 · Result capture	The process is lost mid-run — error kind process_lost	The run log. Resume recovers it; a fresh run is not needed.

WHEN IT MATTERS Session persistence records a session ID before and after each run. If the pair differs the session rotated; if it matches, the conversation continued.

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The Seven Agent States

1. **active**
   Enabled and ready for heartbeats.
2. **idle**
   Active, but no heartbeat is currently running.
3. **running**
   A heartbeat is in progress right now.
4. **error**
   The last heartbeat failed — not working, and did not stop gracefully.
5. **paused**
   Deliberately stopped, by you or automatically on hitting the budget limit.
6. **terminated**
   Permanently deactivated. NOT reversible.
7. **pending_approval**
   Hired but awaiting board approval before it may run.
8. **Run statuses differ**
   Runs are queued, running, succeeded, failed, timed out or cancelled — plus the error kind process_lost.

Pause is reversible. Terminate is not. Read the menu item twice before clicking.

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LO2 · MANAGED BUNDLE OR EXTERNAL VIA ROOTPATH

The Four Instruction Files

File	Answers	Loaded
AGENTS.md	The operating manual — the entry file, and the default.	Every run. Keep it the map, not the whole territory.
SOUL.md	Persona and decision philosophy — how this role weighs trade-offs.	When judgement is needed, not on every wake.
HEARTBEAT.md	The per-heartbeat checklist — what to do on every single wake.	Every run. This is the file that costs you on every heartbeat.
TOOLS.md	Available tools and API notes — what this role may call.	When the agent needs to know its own capability surface.

WHEN IT MATTERS The split matters for cost as much as clarity: material needed on every run is separated from material that is only occasionally relevant.

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Reading the Dashboard at a Glance

Signal	Healthy	Needs attention
Agents enabled	Mostly active or running	Any in error, or unexpectedly paused
Tasks in progress	Work is moving	Nothing in progress for hours, or many blocked
Month spend	Climbing predictably	A sudden jump, or active budget incidents
Pending approvals	Queue manageable	Approvals piling up unattended
Recent activity	Steady, understandable motion	Long silence, or repeated failures

WHEN IT MATTERS The healthy pattern for a small company is ONE agent running while the others stay active — otherwise you cannot attribute a change by timing alone.

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Adapter Type Keys — The Eleven

1. **claude_local**
   Claude Code running locally.
2. **codex_local**
   OpenAI Codex CLI running locally.
3. **gemini_local**
   Gemini CLI running locally.
4. **opencode_local**
   OpenCode running locally.
5. **cursor**
   Cursor — local and cloud variants exist.
6. **pi_local**
   Pi running locally.
7. **hermes_local**
   Hermes running locally.
8. **grok_local**
   Grok running locally.
9. **process**
   An arbitrary local process.
10. **http**
    An HTTP or webhook endpoint.
11. **openclaw_gateway**
    OpenClaw via its gateway.

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Adapter Defaults That Cost You Money

1

maxTurnsPerRun · 300

The Claude Code adapter defaults to a maximum of 300 turns per run. That is a lot of tokens if a loop goes wrong.

2

dangerouslySkipPermissions · true

The Claude Code adapter default. Know it is set before you let an agent run unattended.

3

Sandbox timeout · 14,400s

Four hours. A stuck run holds the workspace for that long unless you intervene.

4

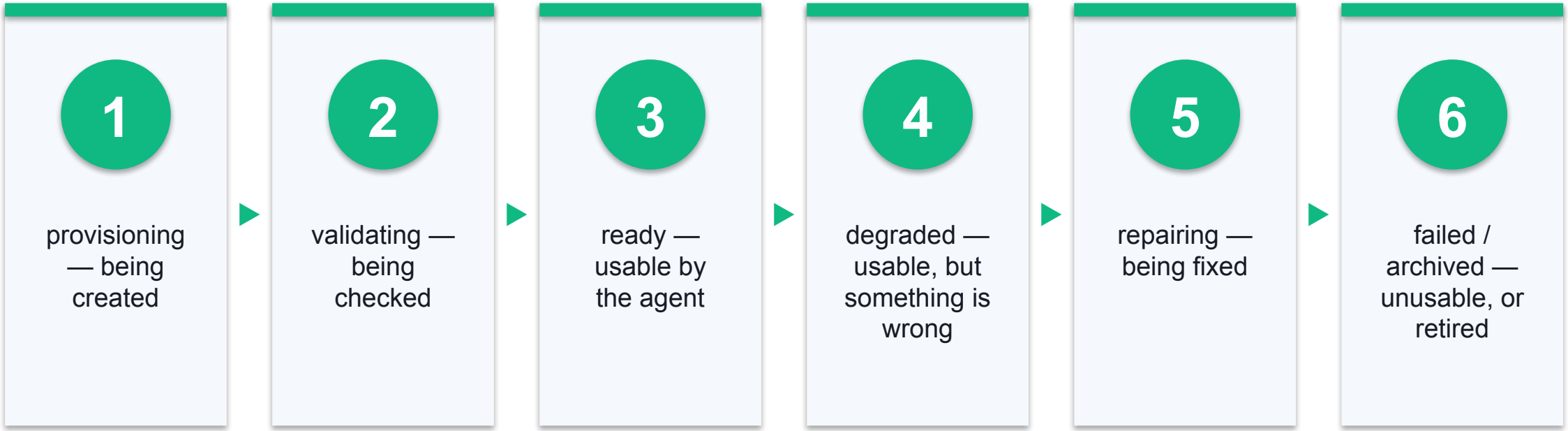
Look before you leave it

Every one of these is a number you should read during configuration, not after an incident.

Adapter readiness is proved by a marker run — never by a saved configuration.

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The Workspace Lifecycle



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LO2 · CONTEXT ECONOMY

Skills — Capability on Demand, Not in Every Prompt

A skill is a reusable instruction document an agent loads only when relevant. The economy is the point: ten skills, but not ten skills loaded per heartbeat.

```
skills/
└── code-review/
    ├── SKILL.md
    └── references/
        └── examples.md

# SKILL.md frontmatter
---
name: code-review
description: >
  Use when asked to review a pull
  request or code diff.
  Don't use when writing new code
  from scratch.
---
```

DESCRIPTION IS DECISION LOGIC

Write when to use it AND when not to — that sentence is what the agent reads to decide whether to load the skill at all.

COMPANY-SCOPED

Skills are never shared across companies. Folders may nest up to four levels deep.

ATTACHED AT HIRE OR LATER

Via desiredSkills at hire time, or later through the skills sync endpoint with mode add, remove or replace.

A DOCUMENTED CONTRADICTION

The Skills API says bundled skills are force-unioned into every agent; the newer CLI page says they are not. Verify on your own install.

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WHAT YOU'LL DO

Hands-On Labs — Building AI Agents with Paperclip

Lab 2

- Create the BrightFunnel Company, Mission and CEO (Digital Marketing)
- BrightFunnel Marketing (Digital Marketing)

Lab 3

- Build the Meridian Reporting Tree and Validate Runtime Boundaries (Finance)
- Meridian Finance Desk (Finance)

—

- —

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LAB 2 · HANDS-ON

Create the BrightFunnel Company, Mission and CEO (Digital Marketing)

SIMULATED COMPANY · BrightFunnel Marketing (Digital Marketing)

You stand up BrightFunnel Marketing — a one-person digital marketing company — as a Paperclip company. You write a MEASURABLE mission (a target, a metric and a boundary), hire the Marketing Director as CEO, connect one trainer-approved agent runtime, and set the company budget cap BEFORE any agent is allowed to run. The order matters: a company that gains a runtime before it gains a cap can spend without a ceiling.

YOU'LL BUILD

A BrightFunnel company in Paperclip with the exact synthetic mission, a hardened Marketing Director CEO on a verified adapter, a saved company budget cap, a dedicated workspace and a redacted configuration baseline in evidence/.

MAPS TO

LO2: Create a goal-aligned Paperclip company with a measurable mission, a hardened CEO agent and enforced company-level caps.

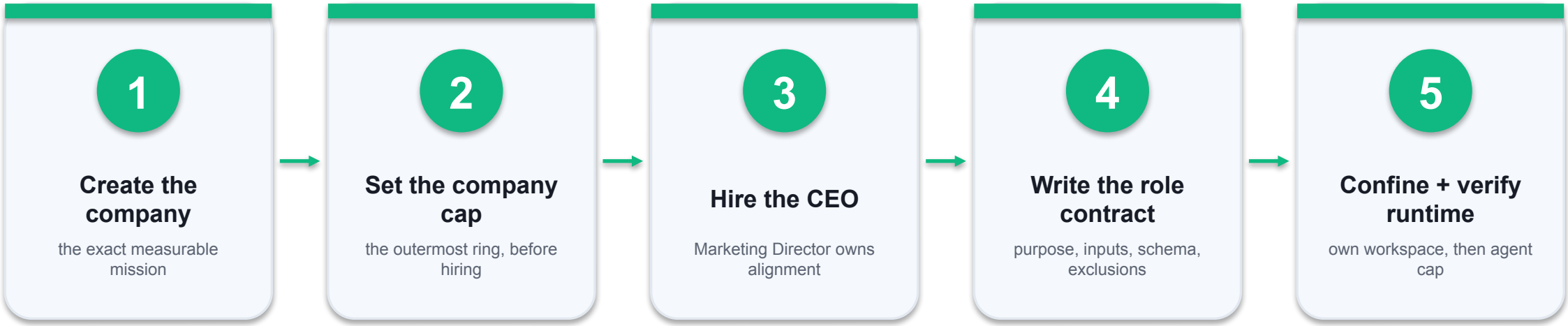
Tools: Paperclip onboarding wizard, Agent Configuration, one approved adapter runtime, text editor

Full step-by-step procedure: Learner Guide, Lab 2 · labs/ folder

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LAB 2 · PROCESS MAP

How Lab 2 Runs



THE POINT

LO2: Create a goal-aligned Paperclip company with a measurable mission, a hardened CEO agent and enforced company-level caps.

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LAB 2 · VERIFY

Create the BrightFunnel Company, Mission and CEO (Digital Marketing)

✅ Expected result

BrightFunnel exists with the exact mission text; the Marketing Director is the sole top-level agent with a verified adapter, a confined workspace and an instruction file; the agent cap (SGD 15) is below the company cap (SGD 40); and the baseline note contains no key, token or connection string.

IF IT DOESN'T WORK

The adapter verification fails

Confirm the underlying CLI runtime is installed and authenticated OUTSIDE Paperclip first, then re-verify. Paperclip cannot authenticate a runtime for you.

The mission field rejects the text

Remove smart quotes and line breaks pasted from a word processor; enter the mission as plain text.

The budget cap will not save

Set the company cap before the agent cap — an agent cap above a missing or smaller company cap is rejected.

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LAB 2 · TROUBLESHOOTING

Create the BrightFunnel Company, Mission and CEO (Digital Marketing)

⚠ The adapter verification fails

Confirm the underlying CLI runtime is installed and authenticated OUTSIDE Paperclip first, then re-verify. Paperclip cannot authenticate a runtime for you.

⚠ The mission field rejects the text

Remove smart quotes and line breaks pasted from a word processor; enter the mission as plain text.

⚠ The budget cap will not save

Set the company cap before the agent cap — an agent cap above a missing or smaller company cap is rejected.

CHALLENGE

Rewrite the BrightFunnel mission so it would FAIL a feasibility scan, then state in one sentence exactly which property (target, metric or boundary) you removed.

REFLECTION

Why is setting the company budget cap before hiring the CEO a control decision rather than an administrative one?

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LAB 3 · HANDS-ON

Build the Meridian Reporting Tree and Validate Runtime Boundaries (Finance)

SIMULATED COMPANY · Meridian Finance Desk (Finance)

You extend the pattern into Meridian Finance Desk and build a real three-agent org chart: a Finance Lead with an FP&A Analyst and a Treasury Analyst reporting to it. Each specialist receives its OWN workspace, its OWN instruction file and its OWN budget. You then prove every runtime boundary with a marker run — a deliberately trivial task whose only purpose is to produce an observable artifact in the correct workspace. Configuration that has never produced an artifact is not a connection.

YOU'LL BUILD

A three-agent Meridian org chart with separate budgeted workspaces, role instruction files, clean Git baselines and one marker-run artifact per agent proving its runtime boundary.

MAPS TO

LO2: Build a multi-agent reporting hierarchy with distinct, budgeted and confined workspaces, and prove each runtime boundary with a marker run.

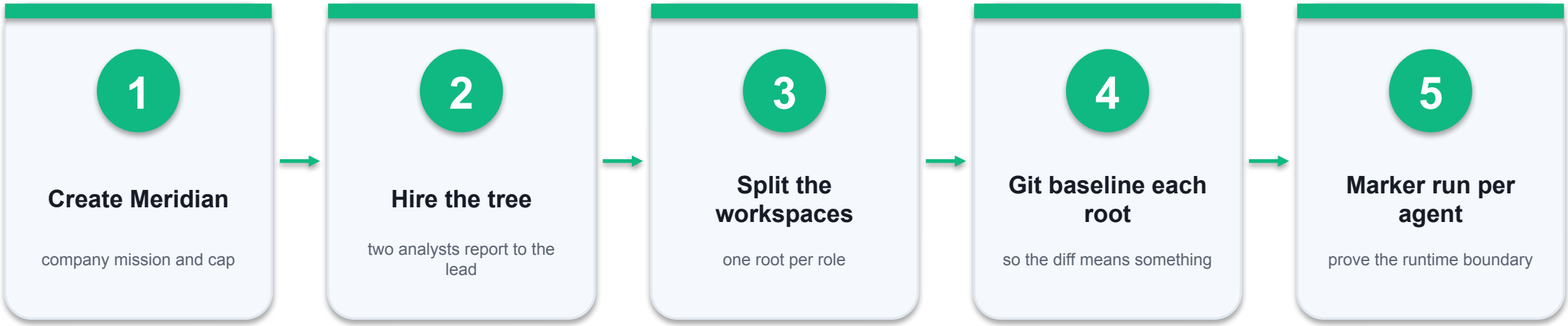
Tools: Paperclip Agents, Org and Runs pages; approved adapter runtime; Git; Markdown editor

Full step-by-step procedure: Learner Guide, Lab 3 · labs/ folder

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LAB 3 · PROCESS MAP

How Lab 3 Runs



THE POINT

LO2: Build a multi-agent reporting hierarchy with distinct, budgeted and confined workspaces, and prove each runtime boundary with a marker run.

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LAB 3 · VERIFY

Build the Meridian Reporting Tree and Validate Runtime Boundaries (Finance)

✅ Expected result

paperclip → Org shows a three-agent tree with both analysts reporting to the Finance Lead; each workspace contains exactly ONE marker.md attributable to its own agent; every agent has an individual cap and the sum of agent caps does not exceed the company cap; and each marker run has a recorded run ID.

IF IT DOESN'T WORK

A marker file appears in the wrong workspace
The agent's workspace path is misbound. Correct the path on the AGENT (not the company), delete the stray file, and re-run the marker.

A specialist never wakes

Confirm the issue is ASSIGNED to that agent — creating an issue does not wake anyone. Then check the agent's adapter verification and its budget headroom.

Git reports the workspace is dirty before the marker run
Something was written outside a run. Reset to the baseline commit and re-run so the diff means what you think it means.

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LAB 3 · TROUBLESHOOTING

Build the Meridian Reporting Tree and Validate Runtime Boundaries (Finance)

⚠ A marker file appears in the wrong workspace

The agent's workspace path is misbound. Correct the path on the AGENT (not the company), delete the stray file, and re-run the marker.

⚠ A specialist never wakes

Confirm the issue is ASSIGNED to that agent — creating an issue does not wake anyone. Then check the agent's adapter verification and its budget headroom.

⚠ Git reports the workspace is dirty before the marker run

Something was written outside a run. Reset to the baseline commit and re-run so the diff means what you think it means.

CHALLENGE

Set the Treasury Analyst's cap to SGD 0.01, re-run its marker task, and record exactly where and how Paperclip reports the budget stop.

REFLECTION

Why is a marker run stronger evidence of a working runtime than a green 'connection verified' badge in the configuration screen?

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TOPIC RECAP

Recap — Building AI Agents with Paperclip

- You can now: LO2: Create a goal-aligned Paperclip company with a measurable mission, a hardened CEO agent and enforced company-level caps.
- You can now: LO2: Build a multi-agent reporting hierarchy with distinct, budgeted and confined workspaces, and prove each runtime boundary with a marker run.
- Onboarding with paperclipai onboard --yes downloads Paperclip, writes the configuration, initialises the embedded PostgreSQL and serves the UI on port 3100.
- Deployment mode is a security decision: local_trusted is the default and requires no login; authenticated+private and authenticated+public widen reach and require real authentication.
- The onboarding wizard captures the company name, the mission statement and the team lead, then verifies connectivity to your chosen local agent runtime.

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03

TOPIC 03

Developing Intelligent Multi-Agent Systems

Projects, goals and lineage · issue decomposition · assignment and hand-off · sequential specialists · verifying between disparate components

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TOPIC WEIGHTING 25%

Key Concepts — Developing Intelligent Multi-Agent Systems

1. Projects hold goals and issues and supply the context an agent reads at runtime, so tasks carry full goal ancestry — the agent sees why, not only what.

2. Goal lineage flows downward: company mission → project goal → parent issue → child issue. An agent that cannot trace its lineage is working without direction.

3. Issues move through seven states — backlog, todo, in_progress, in_review, done, plus blocked and cancelled. done and cancelled are TERMINAL: issues do not move backwards.

4. Decompose into a parent synthesis issue plus specialist child issues. Create every issue unassigned first — assignment is a heartbeat trigger, so an assigned issue wakes an agent immediately.

5. Assignment is the trigger. Assign one child at a time to keep runs sequential, observable and individually attributable.

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TOPIC WEIGHTING 25%

Key Concepts — Developing Intelligent Multi-Agent Systems (continued)

1. An acceptance criterion must name the evidence: which artifact, at which path, containing which fields — never 'the research is done'.

2. A hand-off is accepted only when the receiving agent can consume the artifact without asking the producer a question.

3. Verification between disparate components means tracing one thread across issue ID, run log, comment, workspace diff and cost entry, and finding them consistent.

4. Status is not evidence. An issue marked Done with no linked artifact and no diff is an unverified claim.

5. Explicit limitations are part of a good deliverable — an agent that reports what it could not establish is more useful than one that quietly guesses.

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TOPIC WEIGHTING 25%

Key Concepts — Developing Intelligent Multi-Agent Systems (continued)

1. One event should not create many outcomes: choose a single trigger for the first controlled run and keep the work idempotent.

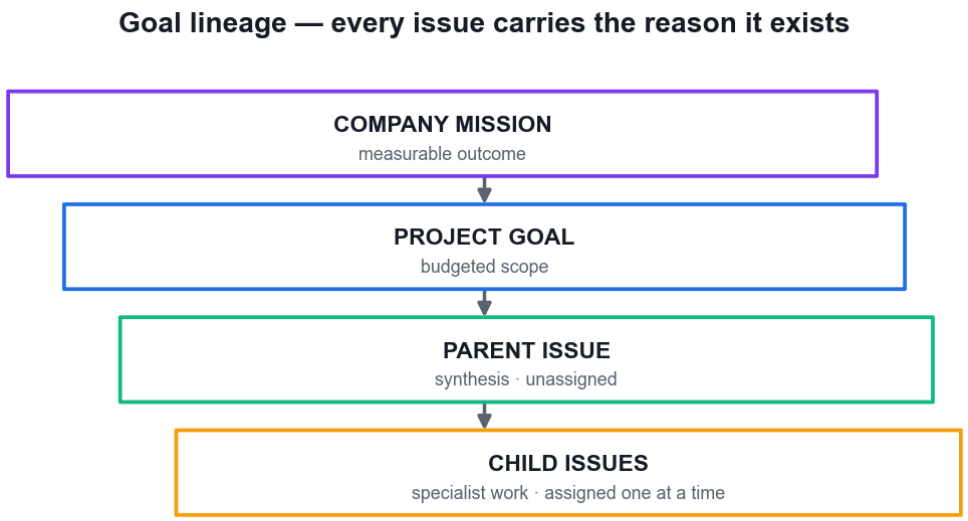
2. Bound every wait. An unbounded wait for a heartbeat hides a failed adapter behind what looks like patience.

3. The workspace diff is the strongest evidence you have — read what actually changed on disk before accepting any agent's summary of it.

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LO3 · THE AGENT SEES WHY, NOT ONLY WHAT

Goal Lineage — Every Issue Carries Its Reason



An agent that cannot trace its lineage is working without direction.

Mission

One measurable outcome defines why the company exists.

Project goal

A budgeted scope the agent reads at runtime.

Parent issue

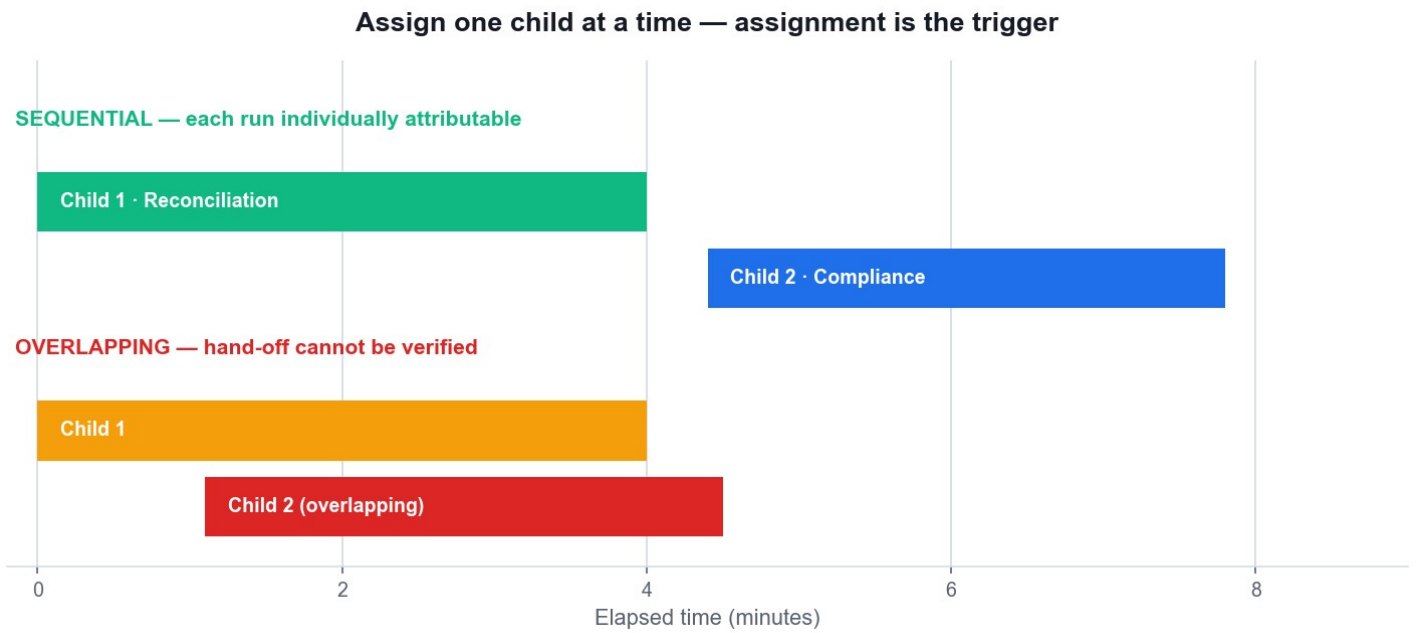
The synthesis — created unassigned so it cannot wake early.

Child issues

Specialist work, assigned ONE AT A TIME.

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Assign One Child at a Time



Assignment is the trigger

Creating an issue does not wake anyone — assigning it does.

Sequential = attributable

Non-overlapping runs let you tie each artifact to exactly one run.

Overlapping = unverifiable

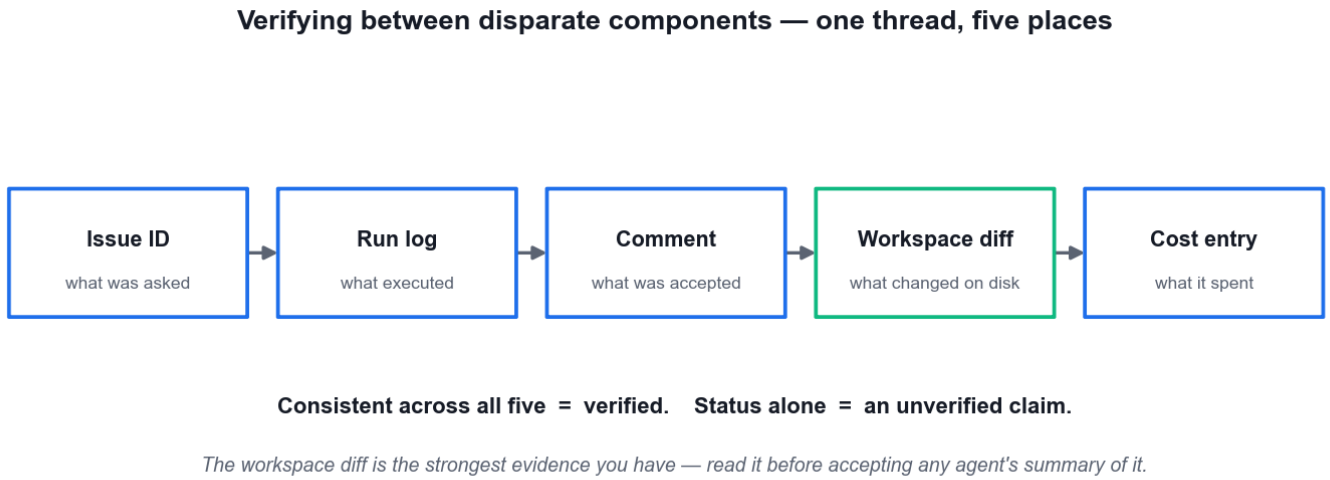
If both ran at once you cannot prove the hand-off actually happened.

Bound every wait

An unbounded wait hides a failed adapter behind apparent patience.

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Verifying Between Disparate Components



Five places, one thread

Issue ID, run log, comment, workspace diff and cost entry must agree.

The diff is king

Read what actually changed on disk before accepting any agent's summary.

Status is not evidence

'Done' with no artifact and no diff is an unverified claim.

Name the evidence

An acceptance criterion states the artifact, its path and its fields.

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Hand-Off Acceptance — What You Are Actually Checking

1

Consumable without a question

The receiving agent can use the artifact without asking the producer anything.

2

Limitations stated

An agent that reports what it could NOT establish is more useful than one that quietly guesses.

3

Idempotent

One event should not create many outcomes — re-running must not duplicate the work.

4

Traceable

Every claim in the deliverable points back to a named source in the allowed inputs.

Accept artifacts and explicit limitations — never a status change on its own.

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LO3 · THE ASSESSMENT EXPECTS THESE BY NAME

The Seven Issue States

State	The documentation's own definition
backlog	The issue exists and has been identified, but no one is working on it yet.
todo	The issue is ready to start. An agent has been assigned and is waiting to check it out.
in_progress	An agent has checked out the issue and is actively working on it. Only ONE agent may hold this state at a time.
in_review	The agent has completed the work and moved the issue to review. It is waiting for sign-off.
done	TERMINAL. The work is complete.
blocked	The agent cannot move forward. Something is preventing progress.
cancelled	TERMINAL. The issue is no longer needed and will not be completed.

WHEN IT MATTERS Issues do not move backwards. A terminal issue is revived only through an explicit reopen — never by editing its status.

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The Normal Issue Path

1. backlog — identified, nobody working
2. todo — assigned, waiting to check out
3. in_progress — checked out, exclusive
4. in_review — done, awaiting sign-off
5. done — terminal

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Checkout Discipline — Four Common Mistakes

Mistake	What to do instead
PATCHing status directly to in_progress	Never do this. Use the checkout mechanism — a direct status write bypasses the lock entirely.
Retrying after a 409 Conflict	Treat 409 as TERMINAL. Another agent holds the issue; retrying will not win the race.
Expecting a cancelled blocker to unblock its dependents	Cancelled blockers never auto-unblock dependents. Resolve the dependency explicitly.
Editing a done issue's status to reopen it	Terminal issues reopen only via an explicit reopen flag, not a status edit.
Writing to the whole board from one run	CROSS_ISSUE_INFLUENCE_LIMIT is 20 — one heartbeat may write to at most twenty other issues.

WHEN IT MATTERS Runs identify themselves with a run-id header. That is how a write is attributed to the execution that made it.

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The Five Heartbeat Triggers

1. **Schedule / interval**
   The configured interval ticks — whether or not there is any work waiting.

2. **Task assignment**
   An issue is assigned to the agent. THIS is what makes assignment the real trigger for work.

3. **Comment mention**
   Someone @-mentions the agent in a comment on an issue.

4. **Manual invoke**
   A human presses Run Heartbeat.

5. **Approval resolution**
   An approval the agent was waiting on is finally decided.

6. **Most people name three**
   The two people miss are approval resolution and the @-mention — and both explain 'why did it just wake?'

Heartbeat on interval is OFF by default — and that default is the right one.

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LO3 · TWO THINGS THAT LOOK THE SAME AND ARE NOT

Paused versus Heartbeat-Off

Condition	Wakes on interval	Wakes on assignment, comment, routine, on-demand
Heartbeat off	No	YES — the agent still responds to events
Paused	No	No — a paused agent wakes on nothing at all
Why it matters	An interval costs money on every tick, work or not	Event-driven waking costs nothing when nothing is happening
The recommendation	Interval OFF	Wake-on-demand ON. The interval is a floor, not a guarantee.

WHEN IT MATTERS A busy agent may run MORE often than its interval, because events trigger wakes independently of the clock.

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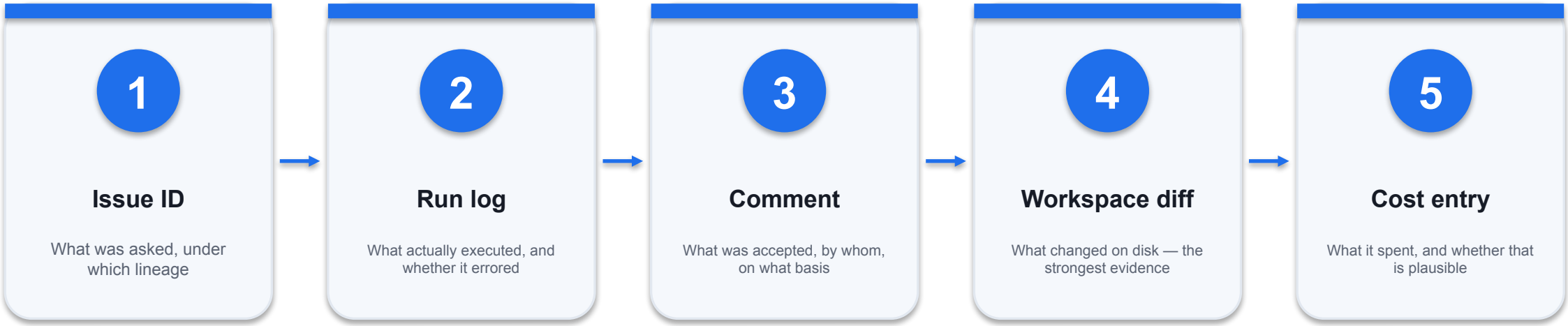
Acceptance Criteria — Activity versus Evidence

Fails — describes activity	Works — names evidence
The research is done.	workspace/research-analyst/research.md exists and contains claim, source, date accessed and confidence for every claim.
Reconcile the control accounts.	reconciliation.md lists every control account with ledger balance, statement balance and variance.
Review for compliance.	exceptions.md lists each variance above SGD 500 with a stated cause and a recommended treatment.

WHEN IT MATTERS The test: could a stranger with no knowledge of the conversation look at the workspace and say yes or no? If not, the criterion is not finished.

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The Evidence Chain — One Thread, Five Places



STATUS IS NOT EVIDENCE

An issue marked done with no linked artifact and no diff is an unverified claim, however confident the agent's summary sounds.

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Execution Policy Parameters

Parameter	Value	What it does
maxReviewRounds	Default 3 · range 1–50	Caps how many times work can bounce back. A null value means the server default of 3.
approvalsNeeded	Fixed at 1	One approval advances the stage. This is not configurable.
changesRequestedCount	Counts review rounds	Escalates to a named reviewer once the cap is reached.
Comment-required backstop	retry_queued → retry_exhausted	The fallback path when a required comment never arrives.

WHEN IT MATTERS Once a reviewer is escalated to, only that reviewer may advance the stage. A 422 'only the escalated reviewer' error is the policy working, not a bug.

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LO3 · SEQUENCE IS A CONTROL, NOT A PREFERENCE

Decomposition and Assignment Discipline

1

Create everything COLD

Parent synthesis issue and all children UNASSIGNED — assignment is a heartbeat trigger.

2

Why unassigned matters

An issue created with an assignee wakes an agent before you have written the acceptance criteria or checked the caps.

3

Assign ONE child at a time

Sequential runs are individually attributable; overlapping runs are not.

4

The proof you lose

If two specialists ran simultaneously you cannot demonstrate the second consumed the first's output.

5

The parent waits

The synthesis issue stays unassigned until every child has produced its artifact.

6

Bound every wait

An unbounded wait hides a failed adapter behind apparent patience.

Creating an issue does not wake anyone. Assigning it does.

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LO3 · LINEAGE IS FOUR LEVELS, NOT ONE SENTENCE

Goal Lineage — What Each Level Supplies

Level	What it carries	What breaks if it is missing
Company mission	The measurable outcome the company exists for	Agents optimise a local goal that does not serve the business.
Project goal	A budgeted scope the agent reads at runtime	Work has no boundary and no lifetime cap.
Parent issue	The synthesis that consumes every child's output	Nobody assembles the pieces; you do it by hand.
Child issue	One specialist transformation with named evidence	The instruction becomes a list of unrelated responsibilities.

WHEN IT MATTERS If you re-explain the goal in every issue body, your lineage is broken — and you are paying for the same context repeatedly.

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LO3 · WHAT YOU ARE ACTUALLY CHECKING

Hand-Off Acceptance Tests

1

Consumable without a question

The receiving agent can use the artifact without asking the producer anything at all.

2

If the receiver must ask

The output schema in the PRODUCER's instruction was under-specified. Fix the contract, not the artifact.

3

Limitations are stated

An agent that reports what it could NOT establish is more useful than one that quietly guesses.

4

Idempotent

One event should not create many outcomes — re-running must not duplicate the work.

5

Traceable

Every claim points back to a named source inside the allowed inputs.

6

The diff is read first

An agent describing its own work is a witness with an interest in the verdict.

Accept artifacts and explicit limitations — never a status change on its own.

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Projects — Lifecycle and the Lifetime Cap

Project state	Meaning	Operator action
backlog	Identified, not yet scheduled	Write the goal before you plan it.
planned	Scoped and budgeted, not started	Confirm the LIFETIME budget cap now — it is distinct from monthly caps.
in_progress	Issues are being worked	Watch the utilisation bar and the run cadence.
completed	The goal has been met	Close the audit map before you archive.
cancelled	Abandoned	Record why. A cancelled project is still evidence.

WHEN IT MATTERS Company and agent budgets are MONTHLY. The project budget is a LIFETIME cap for an execution-bound project — a different period, not a different ring.

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WHAT YOU'LL DO

Hands-On Labs — Developing Intelligent Multi-Agent Systems

Lab 4

- Translate the Ledgerly Mission into Assigned Issues and Hand-Offs (Accounting)
- Ledgerly Accounting (Accounting)

Lab 5

- Verify NorthPoint Routines and Layered Budget Controls (Stock Trading Research)
- NorthPoint Trading Research (Stock Trading — research only)

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LAB 4 · HANDS-ON

Translate the Ledgerly Mission into Assigned Issues and Hand-Offs (Accounting)

SIMULATED COMPANY · Ledgerly Accounting (Accounting)

You turn the Ledgerly Accounting mission into actual work. You create a budgeted month-end close project, then a PARENT synthesis issue and two specialist CHILD issues — all created UNASSIGNED, so that creating work cannot accidentally wake an agent. Only after every acceptance criterion names its evidence do you assign one child at a time. This is the core discipline of the topic: assignment is the trigger, and sequential assignment is what makes each run individually verifiable.

YOU'LL BUILD

A capped Ledgerly close project with an unassigned parent issue, two sequentially completed specialist children, clean workspace manifests, and an evidence record linking issue IDs, run IDs, comments, statuses and artifact paths.

MAPS TO

LO3: Decompose a mission into a parent and specialist child issues, run the specialists sequentially and verify each hand-off against acceptance evidence.

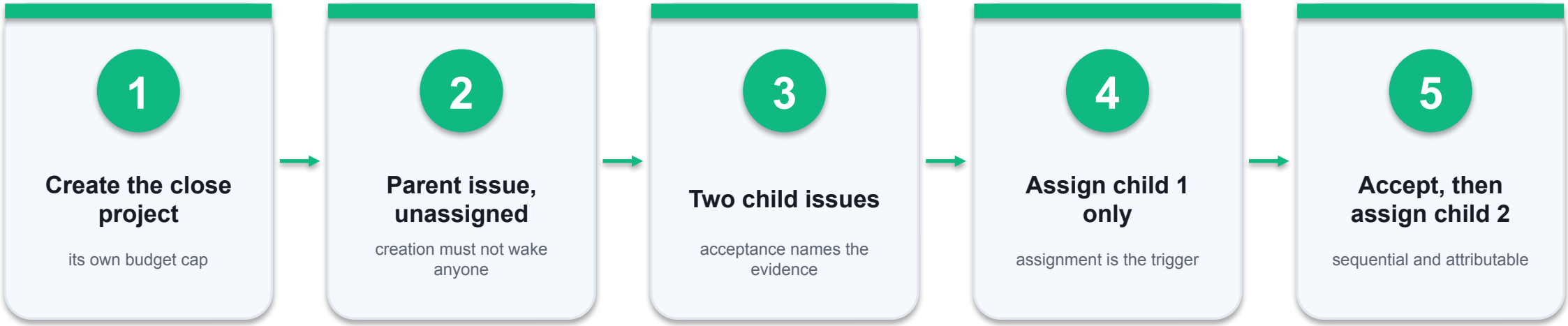
Tools: Paperclip Projects, Budgets, Issues, Agents and Runs; approved adapter runtime; Git; Markdown workspaces

Full step-by-step procedure: Learner Guide, Lab 4 · labs/ folder

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LAB 4 · PROCESS MAP

How Lab 4 Runs



THE POINT

LO3: Decompose a mission into a parent and specialist child issues, run the specialists sequentially and verify each hand-off against acceptance evidence.

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LAB 4 · VERIFY

Translate the Ledgerly Mission into Assigned Issues and Hand-Offs (Accounting)

✅ Expected result

The Ledgerly project shows a saved cap above actual spend; the parent issue is still open and unassigned; both children are Done with a linked artifact and a recorded run ID; the two specialists ran SEQUENTIALLY (non-overlapping run timestamps); and the exceptions file demonstrably references the reconciliation figures.

IF IT DOESN'T WORK

Both specialists wake at the same time
A child was assigned before the previous one was accepted. Pause the second agent, let the first run settle, then re-assign so the runs are sequential.

The child artifact exists but is missing fields
The acceptance criterion was written as an activity, not as evidence. Rewrite it naming the required fields and re-run — do not hand-edit the agent's artifact.

The parent issue wakes unexpectedly
It was assigned during creation. Clear the assignee; the parent stays unassigned until both children are accepted (that is Lab 6).

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LAB 4 · TROUBLESHOOTING

Translate the Ledgerly Mission into Assigned Issues and Hand-Offs (Accounting)

⚠ Both specialists wake at the same time

A child was assigned before the previous one was accepted. Pause the second agent, let the first run settle, then re-assign so the runs are sequential.

⚠ The child artifact exists but is missing fields

The acceptance criterion was written as an activity, not as evidence. Rewrite it naming the required fields and re-run — do not hand-edit the agent's artifact.

⚠ The parent issue wakes unexpectedly

It was assigned during creation. Clear the assignee; the parent stays unassigned until both children are accepted (that is Lab 6).

CHALLENGE

Add a third child issue whose acceptance criterion deliberately names an artifact the previous child never produces, and record precisely how the failure surfaces in the run log.

REFLECTION

Why does creating every issue unassigned change the risk profile of a multi-agent company, compared with assigning at creation time?

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LAB 5 · HANDS-ON

Verify NorthPoint Routines and Layered Budget Controls (Stock Trading Research)

SIMULATED COMPANY · NorthPoint Trading Research (Stock Trading — research only)

NorthPoint Trading Research needs a daily pre-market brief — which makes it the right place to learn routines. You create the readiness routine PAUSED, add an explicit cron trigger with an IANA timezone, prove the routine once by hand, and leave it paused afterwards. You then verify all three budget layers against actual recorded spend. The rule the lab teaches: automation earns the schedule only after a human has watched it succeed once. This is research only — the company never places or recommends a trade.

YOU'LL BUILD

A paused NorthPoint readiness routine with a saved 07:00 Asia/Singapore trigger, exactly one reviewed manual run, and a control table proving company, project and agent budget layers all remain above recorded spend.

MAPS TO

LO3: Test a scheduled routine and verify that every budget layer holds, using one bounded manual run before automation is allowed to own the schedule.

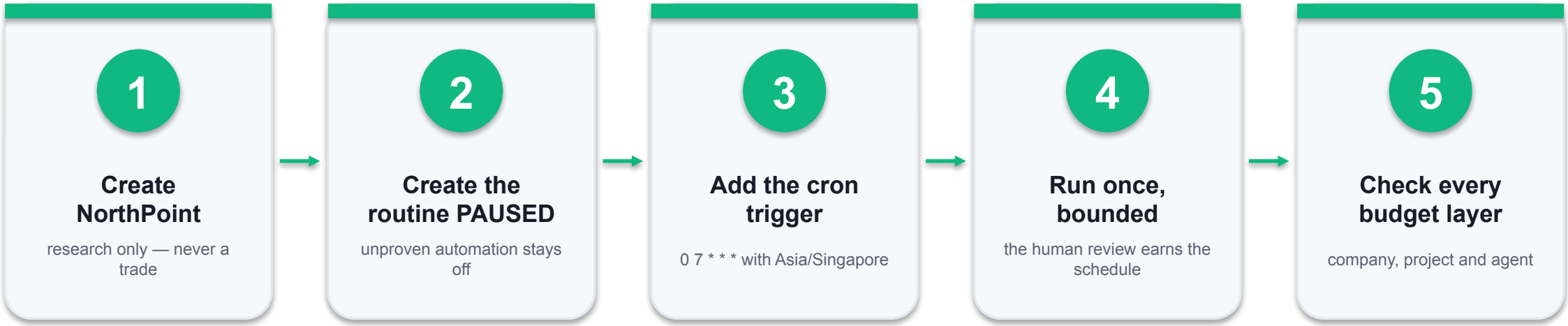
Tools: Paperclip Routines, Costs and Budgets, Agent Budget tabs, Project settings, Activity

Full step-by-step procedure: Learner Guide, Lab 5 · labs/ folder

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LAB 5 · PROCESS MAP

How Lab 5 Runs



THE POINT

LO3: Test a scheduled routine and verify that every budget layer holds, using one bounded manual run before automation is allowed to own the schedule.

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LAB 5 · VERIFY

Verify NorthPoint Routines and Layered Budget Controls (Stock Trading Research)

✅ Expected result

The routine carries an explicit 0 7 * * * Asia/Singapore trigger, has exactly ONE successful reviewed manual run, and finishes PAUSED; the brief exists and cites only the synthetic data with limitations stated and no trade recommendation; and the control table shows all three budget layers above recorded spend.

IF IT DOESN'T WORK

The routine fires unexpectedly

It was left active after the manual run. Pause it, then check the activity log for any additional runs and record them — do not delete the evidence.

The cost shows as zero

The run may have failed before invoking the runtime. Read the run log for an adapter error rather than assuming the work was free.

The timezone reverts or the schedule fires an hour out

The trigger was saved without an explicit IANA timezone and inherited the host's. Re-save the trigger with Asia/Singapore stated explicitly.

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LAB 5 · TROUBLESHOOTING

Verify NorthPoint Routines and Layered Budget Controls (Stock Trading Research)

⚠ The routine fires unexpectedly

It was left active after the manual run. Pause it, then check the activity log for any additional runs and record them — do not delete the evidence.

⚠ The cost shows as zero

The run may have failed before invoking the runtime. Read the run log for an adapter error rather than assuming the work was free.

⚠ The timezone reverts or the schedule fires an hour out

The trigger was saved without an explicit IANA timezone and inherited the host's. Re-save the trigger with Asia/Singapore stated explicitly.

CHALLENGE

Lower the Market Data Analyst cap below the recorded run cost, run the routine once more, and record exactly which layer stops the run and where it is reported.

REFLECTION

Why must a routine be proved by one bounded manual run before the schedule is allowed to own it?

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TOPIC RECAP

Recap — Developing Intelligent Multi-Agent Systems

- You can now: LO3: Decompose a mission into a parent and specialist child issues, run the specialists sequentially and verify each hand-off against acceptance evidence.
- You can now: LO3: Test a scheduled routine and verify that every budget layer holds, using one bounded manual run before automation is allowed to own the schedule.
- Projects hold goals and issues and supply the context an agent reads at runtime, so tasks carry full goal ancestry — the agent sees why, not only what.
- Goal lineage flows downward: company mission → project goal → parent issue → child issue. An agent that cannot trace its lineage is working without direction.
- Issues move through seven states — backlog, todo, in_progress, in_review, done, plus blocked and cancelled. done and cancelled are TERMINAL: issues do not move backwards.

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04

TOPIC 04

Deploying and Managing AI Agents

Routines and schedules · layered budgets · approvals and audit · operator intervention · deployment, portability and troubleshooting

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TOPIC WEIGHTING 25%

Key Concepts — Deploying and Managing AI Agents

1. A routine is recurring work triggered by cron, webhook or API. Create it paused, prove it once by hand, and only then let the schedule own it.

2. A schedule trigger is explicit — 30 16 * * * with an IANA timezone such as Asia/Singapore. An implicit timezone is a production incident waiting to happen.

3. Budgets have three scopes enforced in PARALLEL: company (monthly), agent (monthly) and project (lifetime). An agent can be paused by its own cap even when the company has budget left.

4. Enforcement is a two-stage circuit breaker at exact thresholds: 80% utilisation records a warning and the agent keeps running; 100% pauses the agent automatically with pauseReason = budget. Enforcement happens at checkout, not at billing.

5. Treat the budget as a two-stage circuit breaker — warn, then stop — and record the baseline spend and expected run frequency before you rely on it.

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TOPIC WEIGHTING 25%

Key Concepts — Deploying and Managing AI Agents (continued)

1. Cost is tracked at token level by provider, model, agent and project, which is what makes the per-agent cap meaningful rather than notional.

2. Approvals come in three types — CEO Strategy, Hire Agent and Budget Override — and you have three responses: approve, reject, or request revision. Approvals never expire on their own, so a silent company is often an unattended approval queue.

3. The activity log is a durable record of mutations, heartbeats, cost events and approvals — the audit loop closes when the whole operation reconstructs from it.

4. Use the least irreversible control first: inspect, pause, reassign, then terminate. Pause and heartbeat-off differ: heartbeat-off still wakes on assignment, comment, routine and on-demand; a paused agent wakes on nothing.

5. Deployment modes are local_trusted (the default, no login), authenticated+private and authenticated+public, set with paperclipai configure --section server.

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TOPIC WEIGHTING 25%

Key Concepts — Deploying and Managing AI Agents (continued)

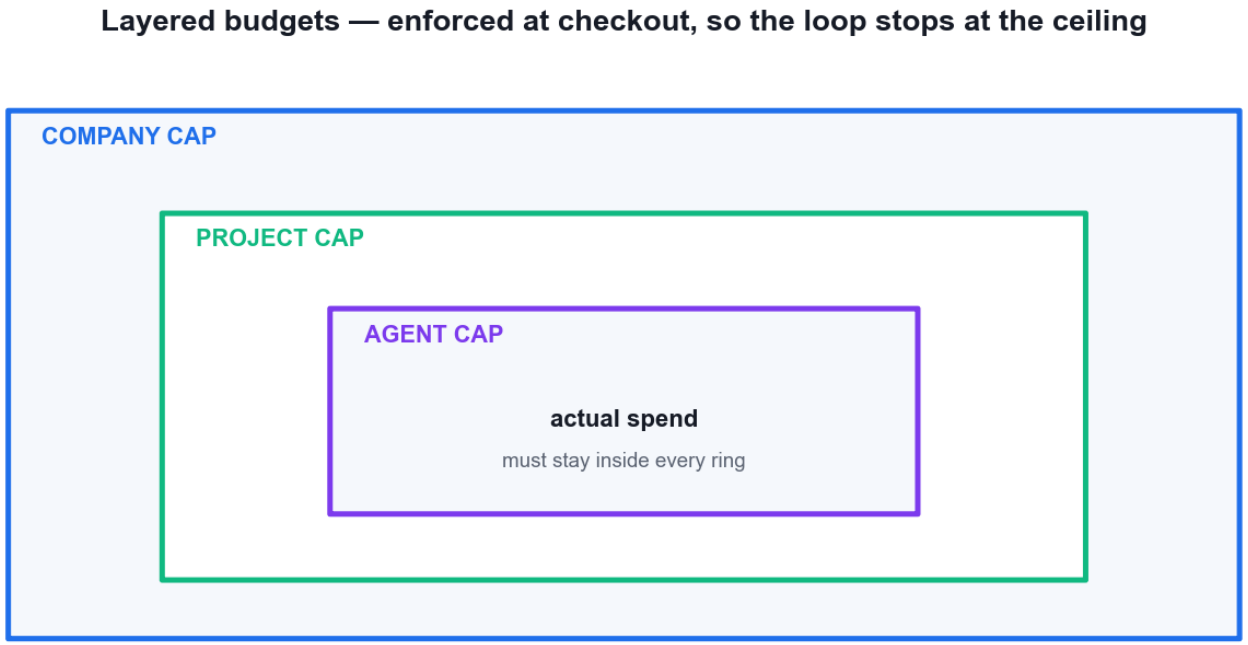
1. A hosted deployment REQUIRES PAPERCLIP_AGENT_JWT_SECRET and DATABASE_URL; PAPERCLIP_TOOL_ACTION_SIGNING_SECRET is separately documented as required with no fallback for TOOL-ACTION approvals.

2. Portability means an entire company exports and imports with secrets scrubbed and collisions handled — the org chart is data, not a hand-built configuration.

3. Troubleshooting is a discipline: reproduce, isolate the layer, change one setting, re-verify with evidence, and record what the fix actually proved.

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Layered Budgets — Warn, Then Stop



Warn, then stop. The innermost cap that binds is the one that fires first.

Three rings

Company, project and agent — the innermost cap that binds fires first.

Enforced at checkout

Not at billing. The loop stops at the ceiling.

Baseline first

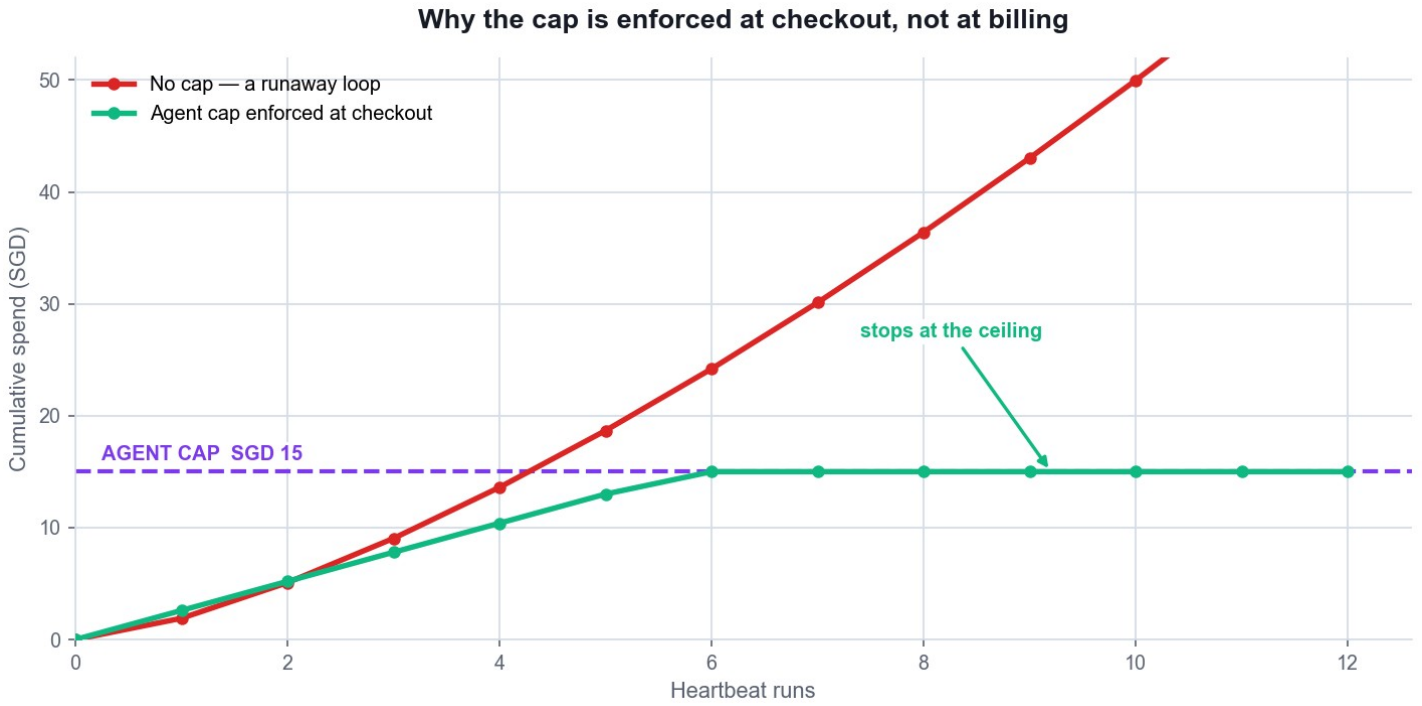
Record current spend and expected run frequency before you rely on a cap.

Token-level costing

Tracked by provider, model, agent and project — which is what makes the cap meaningful.

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Why the Cap Is Enforced at Checkout



Without a cap

A runaway loop keeps spending until someone notices the invoice.

With a cap

Spend flattens at the ceiling and the agent simply stops.

The cap is a control

Not a budget estimate — it is the thing that makes unattended operation safe.

Set it below the pain

A cap you would be unhappy to actually pay is set too high.

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Routines — Automation Earns the Schedule

1

Create it PAUSED

A routine live before it has ever succeeded is unreviewed scheduled spend.

2

State the trigger explicitly

30 16 * * * WITH an IANA timezone such as Asia/Singapore.

3

Prove it once by hand

One bounded manual run, reviewed by a human.

4

Leave it paused

Until you have read the generated issue, the run and the cost.

5

Name the artifact

The routine body must state what file it produces, or the run is unverifiable.

6

Watch for concurrency

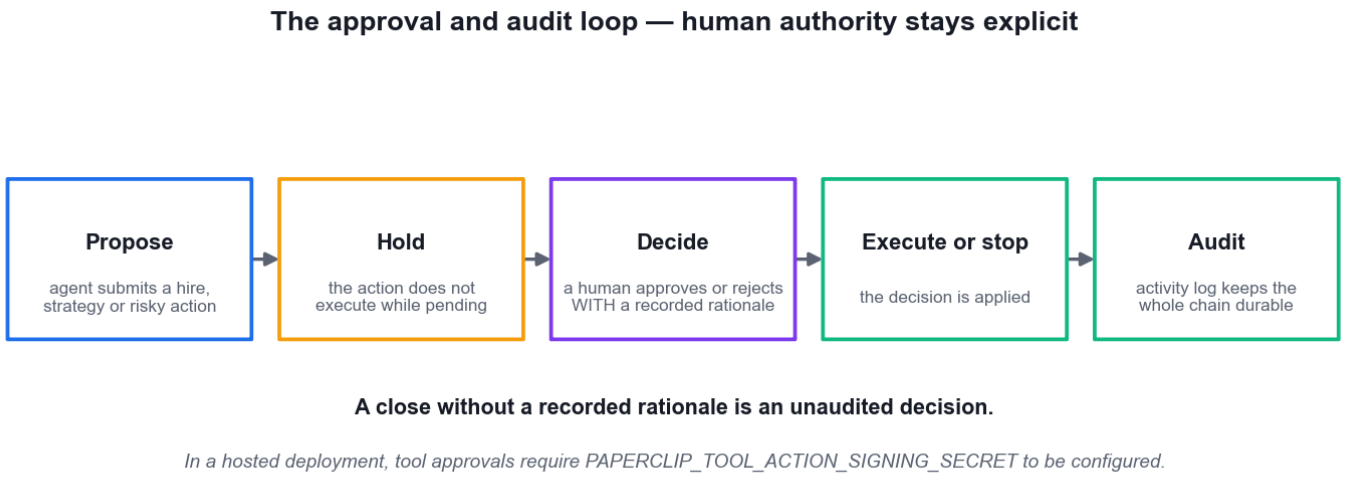
One event should not create many outcomes.

An implicit timezone is a production incident waiting to happen.

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LO4 · HUMAN AUTHORITY STAYS EXPLICIT

The Approval and Audit Loop



Propose, then hold

The governed action does not execute while it is pending.

A human decides

With a RECORDED rationale — that sentence is the audit.

The log is durable

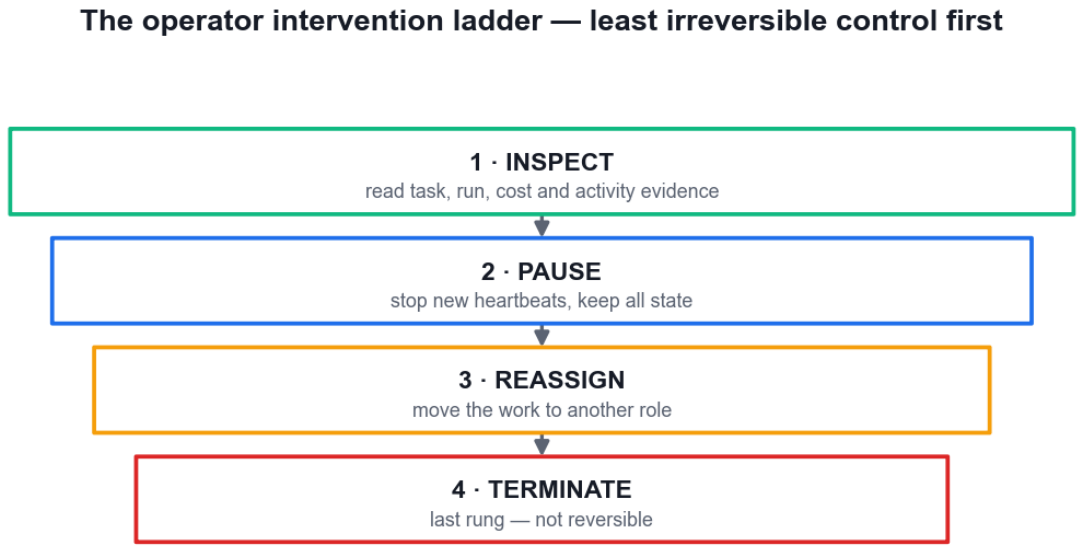
Mutations, heartbeats, cost events and approvals all land in Activity.

The loop closes

When the whole operation can be reconstructed from the log alone.

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The Operator Intervention Ladder



Escalate only when the rung below has genuinely failed — and terminate only a role you created as disposable.

Inspect

Read task, run, cost and activity evidence BEFORE acting.

Pause

Stop new heartbeats while keeping all state.

Reassign

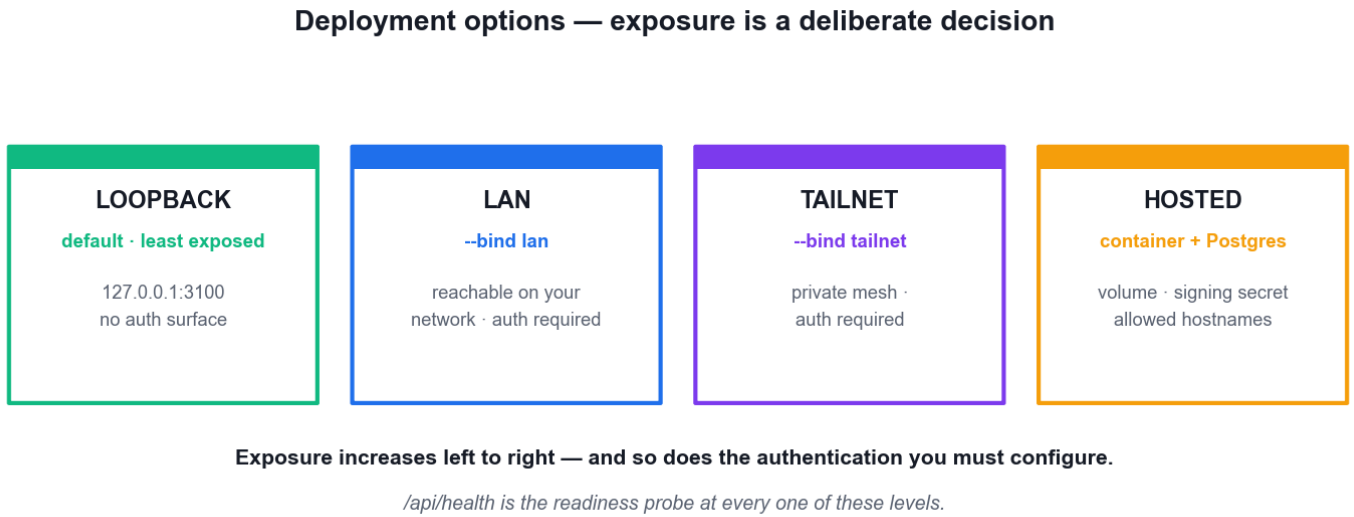
Move the work to another role.

Terminate

The last rung — and it is not reversible.

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Deployment Options — Exposure Is a Decision



Loopback

The default and least exposed — trusted local only.

LAN / tailnet

--bind lan or --bind tailnet widen reach and REQUIRE authentication.

Hosted

Container plus external PostgreSQL, a persistent volume, a signing secret and an allowed-hostnames list.

Portability

A whole company exports and imports with secrets scrubbed and collisions re-mapped.

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LO4 · THE MOST EXPENSIVE MISUNDERSTANDING IN THE COURSE

The Three Budget Scopes — Parallel, Not Nested

Scope	Period	Applies to
Company budget	Monthly	The total ceiling for the entire company.
Per-agent budget	Monthly	One individual agent's ceiling.
Project budget	Lifetime	An execution-bound project, for the whole of its life.
The consequence	They are independent	An agent can be paused by its OWN budget limit even if the company has remaining budget.

WHEN IT MATTERS Do not picture nested rings where the outer one always binds first. Picture three separate trip switches, any one of which can fire.

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Thresholds and Utilisation Colours

1

80% utilisation

A WARNING is recorded so you can intervene. The agent CONTINUES running — nothing stops.

2

100% utilisation

The agent is automatically PAUSED. No further heartbeats fire, and pauseReason is set to budget.

3

Green · below 70%

Read the utilisation bar for company health at a glance.

4

Yellow · 70 to 90%

Look closer. Something is either busier or more expensive than you planned.

5

Red · above 90%

Intervene now — the hard stop at 100% is close.

6

Enforced AT CHECKOUT

Before the work runs, not when the invoice arrives. That is what makes unattended operation safe.

A cap you would be unhappy to actually pay is set too high.

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Inference Spend versus Finance Events

	Inference spend	Finance events
What it records	Per-request model cost	Subscriptions, invoices, credits, refunds, adjustments
Tracked by	Provider, model, agent and project	The commercial relationship, not the run
Drives the cap	YES — this is what utilisation measures	No — a credit does not raise an agent's ceiling
The mistake	Reading one and thinking you have seen the other	The Costs page separates them deliberately

WHEN IT MATTERS Baseline current spend and expected run frequency BEFORE you rely on a cap. A ceiling you have not calibrated is a guess with a number on it.

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The Three Approval Types

1

CEO Strategy

Raised when the CEO produces a strategic plan after its first heartbeat. The CEO cannot move tasks to in-progress until you approve it.

2

Hire Agent

Raised when an agent — usually the CEO or a manager — proposes a subordinate.

3

The hire payload

Name, role, capabilities, adapter configuration, monthly budget and reporting line. Read every field before approving.

4

Budget Override

Raised when a budget policy hits a hard stop and someone asks to proceed anyway. This is the one to slow down on.

5

Approvals never expire

An agent waiting on a pending approval pauses indefinitely. A silent company is very often an unattended queue.

6

The board override

The board may pause any agent and resume any paused agent, bypassing the approval flow entirely.

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LO4 · THE THIRD RESPONSE IS THE USEFUL ONE

The Three Approval Responses

1

APPROVE

The action proceeds. For a hire, the agent is created, configured and queued to wake automatically.

2

REJECT

The action is denied. The requesting agent is notified and will not retry unless instructed.

3

REQUEST REVISION

You are not approving as-is, but you are not saying no. The agent revises and resubmits. This is the one people forget exists.

4

The state machine

pending goes to approved, rejected, or revision_requested — then resubmitted, and back to pending.

5

RECORD THE RATIONALE

One sentence explaining why is the difference between a decision and an unaudited event.

6

Months later

The rationale is the only thing that will tell you — or a regulator — what you were thinking.

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The Routine Safety Sequence

1. **Create it PAUSED**
   Never live before it has ever succeeded
2. **State the trigger**
   Cron expression AND an IANA timezone
3. **Name the artifact**
   Or the run succeeds but is unverifiable
4. **Run once by hand**
   Bounded wait; read issue, run and cost
5. **Only then unpause**
   The schedule now owns reviewed work

WHY THE TIMEZONE IS NOT A FORMALITY

A cron expression without a timezone runs in whatever zone the host happens to be in — fine until the host moves, the clock changes, or someone deploys to another region. Write Asia/Singapore explicitly.

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LO4 · ONLY ONE STATUS ACTUALLY FIRES

Routine States and Defaults

1

active

The only status that actually fires. Everything else is dormant.

2

paused

Created here, and it stays here until you have proved it by hand.

3

archived

Retired. Kept for the audit trail, not for execution.

4

draft · running · succeeded · failed

List-view badges, not statuses — do not confuse a badge with the routine's real state.

5

Five-minute granularity

The routine editor snaps minutes to five-minute increments. Plan the schedule around that.

6

Coalesce if active, skip missed

The concurrency default: a routine already running will not start a second copy, and a missed window is skipped.

A routine stuck in running with nothing happening usually means a lost process.

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LO4 · ESCALATE ONLY WHEN THE RUNG BELOW HAS FAILED

The Intervention Ladder — Least Irreversible First

Rung	Action	Reversible?	Use when
1	INSPECT — read task, run, cost and activity evidence	Yes — nothing changes	Always. Never act before you have looked.
2	PAUSE — stop new heartbeats, keep all state	Yes	The agent is doing something wrong and must stop now.
3	REASSIGN — move the work to another role	Yes	The work is valid but this agent is the wrong owner.
4	TERMINATE — retire the agent	NO	The role should not exist at all. Last resort only.

WHEN IT MATTERS Practise Terminate only on a role you deliberately created as disposable — never as a first response to a misbehaving production agent.

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RBAC — Company Roles and Their Grants

Role	Description	Implicit grants
Owner	The people who run the company	All eight — agents, skills, environments, users, tasks and joins
Admin	Trusted operators who onboard people and agents	As Owner, MINUS users:manage_permissions
Operator	Hands-on members who help run the work	tasks:assign
Viewer	Read-only observers	None
Unset	No implicit grants at all	Hand-picked explicit grants only — the tightest configuration available

WHEN IT MATTERS Owner's eight: agents:create, agents:configure, skills:create, environments:manage, users:invite, users:manage_permissions, tasks:assign, joins:approve. An explicit grant PERSISTS across a role change, and there is NO DENY LAYER — you cannot subtract a permission.

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Deployment Modes — Exposure Is a Decision

Mode	Login	Use it when
local_trusted (the default)	No login required	A single operator on their own machine. This is the mode used throughout this course.
authenticated + private	Login required, not publicly reachable	A small team on a trusted network or a private mesh.
authenticated + public	Login required, publicly reachable	A hosted deployment. Requires the most care: real auth, real secrets, real exposure.

WHEN IT MATTERS Set with configure --section server, or the PAPERCLIP_DEPLOYMENT_MODE environment variable. You inherit the default only if you never decide.

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LO4 · TWO ARE MARKED REQUIRED; A THIRD HAS NO FALLBACK

Environment Variables You Cannot Skip

1

PAPERCLIP_AGENT_JWT_SECRET

Signs agent tokens. REQUIRED — and it doubles as the fallback for the auth secret.

2

DATABASE_URL

The PostgreSQL connection string. REQUIRED.

3

TOOL_ACTION_SIGNING_SECRET

The PAPERCLIP_ prefixed variable that signs tool-action approvals. Documented as required, with NO fallback.

4

PAPERCLIP_ALLOWED_HOSTNAMES

The hostnames the server will answer for. Add your custom domain here.

5

PAPERCLIP_TELEMETRY_DISABLED

Set to 1 to disable telemetry. DO_NOT_TRACK works the same way.

6

Roughly seventy in total

Across nine categories. You do not need to memorise them — you do need to know which are required.

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LO4 · CONCRETELY, WHAT AN AUDITOR WOULD ASK YOU FOR

The Mission-to-Evidence Audit Map

1

Mission claim

One row per claim the company makes about what it achieved.

2

Issue

Which unit of work was supposed to produce it, and under which lineage.

3

Run

Which execution actually ran, and whether it errored.

4

Artifact

The file on disk, at a named path, with the fields you specified.

5

Cost

What it spent — and whether that is plausible for the work claimed.

6

Approval

The human decision, with its recorded rationale.

The loop closes when the whole operation reconstructs from the log alone.

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Reading the Activity Log for Gaps

What you see	What it usually means	What to do
A steady, ordered sequence of runs	The audit loop is closed	Nothing. This is the state you are working towards.
A workspace change with no run	Work was done OUTSIDE a run — typically hand-edited	Record it honestly. The gap is itself an audit finding.
Long silence, no runs and no spend	An unattended approval queue	Open Approvals first. Approvals never expire on their own.
An agent stopped mid-work	The budget hard stop at 100%	Check utilisation and pauseReason. Raise the cap deliberately, or let it stop.

WHEN IT MATTERS Concealing a gap is far worse than having one. An honest note explaining a hand-edit costs you nothing; a silent gap costs you the whole audit.

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Cost Engineering Levers

1

Turn the interval OFF

It is off by default. An interval wakes the agent whether or not there is work, and each wake costs money.

2

Split the instruction files

Only HEARTBEAT.md is needed on every wake. Move occasional material out of the per-run path.

3

Use skills, not bigger prompts

Ten skills does not mean ten skills loaded per heartbeat — the agent loads only what it decides is relevant.

4

Cap at the agent, not just the company

The scopes are parallel, so the agent cap is what stops one runaway role.

5

Read the adapter defaults

300 turns per run and a four-hour sandbox timeout are large numbers to leave unexamined.

6

Keep lineage intact

Re-explaining the goal in every issue body means paying for the same context over and over.

Every lever here reduces tokens per run — which is the only cost that compounds.

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Portability — A Company Is Data, Not a Craft Object

1. **Export the whole company**
   Org chart, projects, issues, routines and configuration move together as one unit.

2. **Secrets are scrubbed**
   Credentials do not travel in the export. They are re-provisioned at the destination.

3. **Collisions are re-mapped**
   Identifier clashes at the destination are resolved on import rather than failing the whole load.

4. **Back up before a risky change**
   The cheapest insurance available before you restructure an org chart.

5. **Clone a proven structure**
   A second business line starts from a shape you have already debugged.

6. **Laptop to hosted**
   This is how a course company becomes a production one — and how you hand it to a colleague.

If you are afraid to touch a configuration, it is not portable — it is a liability.

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WHAT YOU'LL DO

Hands-On Labs — Deploying and Managing AI Agents

Lab 6

- Run the Governed TalentBridge Operation and Close the Audit Loop (HR)
- TalentBridge People Ops (HR — sourcing, interview, policy, onboarding)

—

- —

—

- —

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LAB 6 · HANDS-ON

Run the Governed TalentBridge Operation and Close the Audit Loop (HR)

SIMULATED COMPANY · TalentBridge People Ops (HR — sourcing, interview, policy, onboarding)

TalentBridge People Ops is the highest-stakes simulation in the course: hiring decisions affect real people, so every control you have learned is now mandatory rather than advisable. You complete the sourcing-to-onboarding operation, review the final synthesis against a workspace DIFF before the board closes it, process a bounded hire approval with a recorded rationale, practise the intervention ladder on a disposable role only, and finally reconstruct the entire operation from the activity log. The audit loop closes when the log alone tells the whole story.

YOU'LL BUILD

A diff-reviewed TalentBridge hiring brief, a separately confined reviewer with a recorded approval rationale, a lifecycle intervention record, and a mission-to-evidence audit map reconstructed from the activity log alone.

MAPS TO

LO4: Resolve issues and modify a running agent company through approvals, the operator intervention ladder and a reconstructed audit trail.

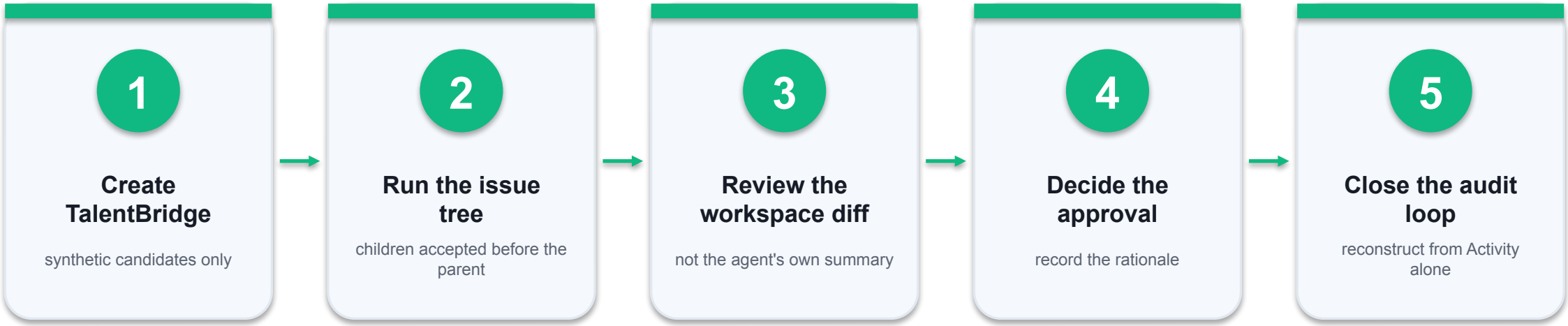
Tools: Paperclip Issues, Agents, Runs, Approvals, Activity, Costs and local Markdown artifacts

Full step-by-step procedure: Learner Guide, Lab 6 · labs/ folder

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LAB 6 · PROCESS MAP

How Lab 6 Runs



THE POINT

LO4: Resolve issues and modify a running agent company through approvals, the operator intervention ladder and a reconstructed audit trail.

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LAB 6 · VERIFY

Run the Governed TalentBridge Operation and Close the Audit Loop (HR)

✅ Expected result

The hiring brief is the only later People Ops Lead workspace change and its diff passes the rubric-traceability and fairness checks; the approval carries a recorded rationale; the intervention ladder was applied to the disposable role only and in order; and the audit map reconstructs every mission claim from activity-log entries with no gap.

IF IT DOESN'T WORK

The CEO synthesises incomplete work
The parent was assigned before both children were accepted. Reopen the parent, accept the outstanding child on named evidence, then re-assign.

The approval cannot be decided
Approvals do not expire on their own — an undecided approval simply waits, and any agent blocked on it waits too. Open Approvals and approve, reject or request revision.

Terminate removes more than intended
Terminate is the LAST rung and is not reversible. Pause and reassign first; only terminate the role you deliberately created as disposable.

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Run the Governed TalentBridge Operation and Close the Audit Loop (HR)

⚠ The CEO synthesises incomplete work

The parent was assigned before both children were accepted. Reopen the parent, accept the outstanding child on named evidence, then re-assign.

⚠ The approval cannot be decided

Approvals do not expire on their own — an undecided approval simply waits, and any agent blocked on it waits too. Open Approvals and approve, reject or request revision.

⚠ Terminate removes more than intended

Terminate is the LAST rung and is not reversible. Pause and reassign first; only terminate the role you deliberately created as disposable.

⚠ The activity log has a gap

Work was done outside a run — for example hand-edited in the workspace. Note it honestly in the audit map; an unexplained gap is itself an audit finding.

CHALLENGE

Export the TalentBridge company, confirm the export scrubs secrets, then import it as a second company and record which identifiers Paperclip had to re-map on collision.

REFLECTION

Your audit map is the deliverable a regulator would actually ask for. Which single row in it is the weakest, and what evidence would strengthen it?

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TOPIC RECAP

Recap — Deploying and Managing AI Agents

- You can now: LO4: Resolve issues and modify a running agent company through approvals, the operator intervention ladder and a reconstructed audit trail.
- A routine is recurring work triggered by cron, webhook or API. Create it paused, prove it once by hand, and only then let the schedule own it.
- A schedule trigger is explicit — 30 16 * * * with an IANA timezone such as Asia/Singapore. An implicit timezone is a production incident waiting to happen.
- Budgets have three scopes enforced in PARALLEL: company (monthly), agent (monthly) and project (lifetime). An agent can be paused by its own cap even when the company has budget left.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1 · Identify & scan feasibility

Explain the Paperclip control-plane model and judge when a managed multi-agent company earns its coordination cost.

2

LO2 · Integrate & develop

Install Paperclip, create a goal-aligned company, build the reporting tree and connect agent runtimes, skills and workspaces safely.

3

LO3 · Test & verify

Decompose a mission into issues, run bounded heartbeats and verify hand-offs through runs, comments, artifacts and the activity log.

4

LO4 · Resolve & modify

Diagnose adapter, budget and routine failures, and modify governance, approvals and cost boundaries to keep the operation safe.

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TAKE THIS BACK TO WORK

A Governed Multi-Agent Operating Checklist

1. **Mission is measurable**
   A target, a metric and a boundary — or it cannot be verified.

2. **Caps exist before runtimes**
   Company, then project, then agent — set before anything runs.

3. **Every role is confined**
   Its own workspace, instruction file, adapter and budget.

4. **Issues are created unassigned**
   Assignment is the trigger; assign one at a time.

5. **Acceptance names the evidence**
   The artifact, its path and its required fields.

6. **The audit reconstructs**
   Activity alone tells the whole story, with no unexplained gap.

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NEXT STEPS

Continue Your Learning

1

Official project

https://github.com/paperclipai/paperclip — the source, the docs and the release notes.

2

Hosted Paperclip

https://paperclip.inc — the managed control plane if you do not want to self-host.

3

Rebuild the labs

Redo each of the six companies from scratch until the control sequence is automatic.

4

Start with one real workflow

Choose an internal task whose inputs and outputs are already observable.

5

Keep the caps low

Your first real company should be capped far below what you would be willing to pay.

6

Audit before you scale

Prove you can reconstruct one operation before you run two.

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CONTINUE WITH TERTIARY INFOTECH

Recommended Courses

1. **WSQ - Develop Multi AI Agent Applications with Gemini Agent ADK**
2. **WSQ - Build LLM Applications Using Flowise and LangChain**
3. **WSQ - Generative AI for Productivity**
4. **WSQ - Prompt Engineering for Business**

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WE'RE HERE TO HELP — DURING AND AFTER THE CLASS

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5

Practice exam
https://exams.tertiaryinfotech.com

6

Course page
www.tertiarycourses.com.sg/wsq-manage-ai-agents-with-paperclip.html

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BEFORE YOU START

Briefing for Assessment

1

Do · Clear your desk

Place phones and other materials under the table or on the floor.

2

Don't · No recording

No photos or recording of assessment scripts.

3

Don't · No discussion

Work individually — no discussion during the assessment.

4

Do · Black or blue pen

Use a black or blue pen for hard-copy assessments.

5

Don't · No correction fluid

No liquid paper or correction tape may be used.

6

Do · Stop on time

Scripts are collected when time is up.

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WRAP-UP

Assessment

1

Written Assessment (SAQ)

5 short-answer knowledge questions · 1 hour · open book.

2

Practical Performance (PP)

4 hands-on Paperclip tasks · 1 hour · open book.

3

Digital attendance

Remember to take the Assessment digital attendance before you start.

4

Submit on the LMS

Upload your completed answers at https://lms-tms.tertiaryinfotech.com/

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ON ASSESSMENT DAY

Assessment Flow

1

TRAQOM survey

Scan the QR code on the LMS

2

Digital attendance

Scan the SSG assessment QR

3

Sit WA then PP

Open book · 1 hour each

4

Submit on the LMS

Upload your completed answers

5

Sign the record

Sign the Assessment Summary Record

REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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HAPPY BUILDING

Thank You!

You can now build, coordinate and govern a Paperclip AI agent company from mission to audited outcome.

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